



**LAKBAY BAYANI: A 2D PLATFORMER GAME PROMOTING PHILIPPINE
HEROISM THROUGH INTERACTIVE LEARNING WITH AUGMENTED
REALITY (AR) INTEGRATION OF NATIONAL HEROES**

**Christian Bacus
Nichole L. Mangopot
Roshelle Masanguid**

**A BSIT Capstone Project Submitted to the Faculty of Computing,
Engineering, and Technology of the Davao Oriental State University in Partial
Fulfillment of the Requirements for the Degree**

BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY

JANUARY 2026

Academic Integrity Declaration

Approval Sheet

ACKNOWLEDGEMENT

The proponents humbly extend their deepest gratitude to every individual and institution who became part of this journey, those who guided, supported, and believed in them from the very first step to the final completion of this capstone project, “Lakbay Bayani: A 2D Platformer Game Promoting Philippine Heroism through Interactive Learning with Augmented Reality (AR) Integration of National Heroes.”

Above all, the proponents offer their sincerest thanks to God, whose wisdom, protection, and unwavering grace sustained them through sleepless nights, unexpected challenges, and moments of doubt. His guidance strengthened their hearts and reminded them that every hardship has a purpose and every effort is meaningful.

The proponents also express profound gratitude to their adviser, Mr. Rich Anjo Capiluyan, whose patience, wisdom, and steady guidance shaped not only the direction of this paper but also their growth as researchers. His constructive feedback and genuine concern inspired them to strive for excellence. Their heartfelt appreciation is also extended to their course coordinator, Ms. Lanie Laureano, for ensuring that they remained aligned with academic standards and for her constant support throughout the process.

To the panel chair, Mr. Clint Dueñas, panel members, Mrs. April Joy Uy, and Mr. Dave Malinab, the proponents offer their utmost thanks. Your insightful comments, thoughtful recommendations, and thorough evaluation strengthened this project in ways words can hardly capture.

The proponents likewise acknowledge the invaluable expertise shared by their subject matter experts, Mr. Vinz Alferez and Mr. Alan Karl Gatdula. Your professional insights added depth and credibility to this study. Their gratitude also goes to Mr. Rodrigo Salimaco, their statistician, whose skillful handling of data ensured the accuracy, integrity, and reliability of the project's evaluation.

With full hearts, the proponents extend their deepest appreciation to their parents, whose unconditional love, sacrifices, and unwavering belief became their foundation. Your moral strength, patience, and continuous support both emotional and financial which gave them courage when they felt weak and motivation when the process seemed impossible. This accomplishment is as much yours as it is theirs.

They also express warm thanks to their friends, Keanu James, Shena Claire, Gabriel, Jenny Babe, and Krystal, who stood beside them through the most challenging stages of research and development. Your laughter, encouragement, and constant reminders to keep going brought light to the most difficult days.

Lastly, the proponents convey their heartfelt appreciation to every individual who, in ways big or small, directly or indirectly contributed to the success of this project. Lakbay Bayani is not merely the product of their efforts, but a collective achievement that built on shared support, guidance, and kindness.

TABLE OF CONTENTS

TITLE	PAGE
Title Page	i
Academic Integrity Declaration	ii
Approval Sheet.....	iii
Acknowledgement.....	iv
Table of Contents.....	vi
List of Tables.....	ix
List of Figures.....	x
List of Appendices.....	x
Abstract.....	xii

CHAPTER I – INTRODUCTION

1.1 Rationale of the Study	1
1.2 Purpose and Project Description	2
1.3 Objectives of the Study	4
1.4 Significance of the Study	5
1.5 Scope and Limitations	6
1.6 Conceptual Framework	9
1.7 Definition of Terms	10

CHAPTER II – REVIEW OF RELATED LITERATURE

2.1 Literature Map	13
2.2 Related Literature.....	14

Game Based Learning in Education	14
Integration of History and Technology in Learning	16
2D Platformer Games and Their Potential in Education	17
Philippine Heroism and Values	18
Integration of Augmented Reality in Learning	19
2.3 Synthesis	20
2.4 Related Systems	21
Bayani	21
Timeline of Jose Rizal and Lapu-lapu	22
Gala Game	23
Monument Valley Game	24
Pokemon GO Game	24

CHAPTER III – MATERIALS AND METHODS

3.1 Software Methodology.....	27
3.2 System Planning	28
Feasibility Study	29
Work Breakdown Structure	32
Gantt Chart.....	33
3.3 System Analysis	34
Description of the Current System	34
Requirement Specification	36
Proposed Solution	38
3.4 System Design	38
System Architecture.....	39

Entity-Relationship Diagram	41
Data Dictionary	41
Data Flow Diagram	45
Use Case Diagram	48
UX Sitemap	54
3.5 System Implementation	55
System Development.....	56
System Testing.....	59
System Deployment.....	61
3.6 System Maintenance.....	62
CHAPTER IV – RESULTS AND DISCUSSIONS	63
CHAPTER V – SUMMARY, CONCLUSIONS, & RECOMMENDATIONS	78
5.1 Summary.....	62
5.2 Conclusions.....	62
5.3 Recommendations	62
LITERATURE CITED	81
APPENDICES	81
CURRICULUM VITAE.....	81

LIST OF TABLES

TABLES	PAGE
Table 2.1 Comparison Matrix of Related Systems	25
Table 3.1 Cost/Benefit Analysis	30
Table 3.2 Data Dictionary	42
Table 3.3 Login Use Case	49
Table 3.4 Register Use Case.....	50
Table 3.5 Play Game Use Case	50
Table 3.6 View Hero 3D Model	51
Table 3.7 Play Level Use case	51
Table 3.8 Check Hero Unlock Status Use Case	52
Table 3.9 Select Hero Use Case	52
Table 3.10 Collect Coin	53
Table 3.11 Collect Artifact Use Case	53
Table 3.12 Unlock & Purchase Hero.....	53
Table 4.1 User Evaluation Results.....	74
Table 4.2 Overall Quality Evaluation of the Lakbay Bayani.....	77

LIST OF FIGURES

FIGURES	PAGE
Figure 1.1 Conceptual Framework.....	9
Figure 2.1 Literature Map	13
Figure 2.2 Bayani: Kanino ka Kakampi?	21
Figure 2.3 Timeline of Jose Rizal and Lapu-Lapu.....	22
Figure 2.4 Gala Game	23
Figure 2.5 Monument Valley Game	24
Figure 2.6 Pokemon GO	24
Figure 3.1 Scrum Framework	27
Figure 3.2 Work Breakdown Structure	32
Figure 3.3 Gantt Chart.....	33
Figure 3.4 Business Process Modelling Notations	35
Figure 3.5 System Architecture	39
Figure 3.6 Entity Relationship Diagram.....	40
Figure 3.7 Data Flow Diagram Level 0.....	45
Figure 3.8 Data Flow Diagram Level 1.....	46
Figure 3.9 Use Case Diagram	48
Figure 3.10 UX Sitemap	54
Figure 4.1 Select Hero	63
Figure 4.2 10 Playable Levels	63
Figure 4.3 Gameplay of Lapu-Lapu	64
Figure 4.4 Gameplay of Jose Rizal.....	65

Figure 4.5 Gameplay of Andres Bonifacio	65
Figure 4.6 Gameplay of Emilio Aguinaldo.....	66
Figure 4.7 Gameplay of Apolinario Mabini	66
Figure 4.8 Cutscenes	66
Figure 4.9 Artifact Collected	66
Figure 4.10 Hero's Information	66
Figure 4.11 Artifacts Collected in Hero Archive.....	66
Figure 4.12 Coins to Collect	66
Figure 4.13 Coins Collected	66
Figure 4.14 Hero Purchasing.....	66
Figure 4.15 Hero Unlocking	66
Figure 4.16 Hero 3D Model in AR.....	66

ABSTRACT

Christian Bacus, Nichole L. Mangopot, and Roshelle Masanguid, “**Lakbay Bayani: A 2D Platformer Game Promoting Philippine Heroism Through Interactive Learning with Augmented Reality (AR) Integration of National Heroes**” (BSIT Capstone Project), Davao Oriental State University 2025

Adviser: Rich Anjo M. Capiloyan

Many learners struggle to retain information about Philippine national heroes due to traditional, text-heavy instructional methods that often fail to sustain engagement. Addressing this issue required exploring interactive digital approaches that could present historical content in a more accessible and appealing format. The study entitled “Lakbay Bayani: A 2D Platformer Game Promoting Philippine Heroism through Interactive Learning with Augmented Reality (AR) Integration of National Heroes” aimed to design, develop, and evaluate a mobile-based educational game that promoted Philippine heroism through interactive gameplay and AR features. The system was developed using the Unity engine and C# to implement 2D platforming mechanics, with an SQLite database managing player progress, scores, and hero unlocks. Levels were inspired by selected national heroes and arranged with increasing difficulty. The study focused on designing the game structure and mechanics, developing the application with integrated AR components, and evaluating the system based on the ISO/IEC 25010 software quality model, specifically functional suitability, performance efficiency, reliability, usability, and security. Usability testing was conducted with fifty (50) respondents selected through quota sampling, who assessed the system using a standardized questionnaire aligned with the identified quality characteristics. The results yielded an overall mean of 4.16 with a descriptive rating of “With High Quality,” indicating that the game met the expected software quality criteria. These findings showed that the system functioned effectively as a game application and satisfied key quality attributes, supporting its potential as a tool for presenting historical content and promoting awareness of Philippine heroism.

Keywords: lakbay bayani, 2d platformer, augmented reality, philippine heroism, educational game

CHAPTER 1

INTRODUCTION

1.1 Rationale of the Study

Education has long contributed to shaping national identity, particularly through lessons on history and heroism. However, many Filipinos continue to struggle with recalling national heroes and their contributions due to traditional teaching methods that rely mainly on textbooks and lectures. These approaches often fail to sustain student interest, as noted by Ebro (2022), who described them as unengaging. Similarly, Calipay (2024) reported that many students could not remember key details about national heroes, reflecting weak retention of historical content.

To address these challenges, educators implemented Game-Based Learning (GBL) as a strategy to make learning more engaging. García-Martínez et al. (2023) found that GBL increased motivation and critical thinking, while Centeno-Martín et al. (2023) emphasized that games dealing with historical topics fostered emotional and cultural bonds. Despite these benefits, most educational games in the Philippines did not focus on local history or align with the national curriculum.

Another promising tool was Augmented Reality (AR), which enriched learning through interactive visuals. Sinha (2023) highlighted AR's ability to engage learners, yet its application in Philippine schools particularly in history classes remained limited. Garcia (2020) demonstrated AR's potential through an app on American colonization, which improved student motivation and comprehension, but similar projects centered on Filipino heroes were scarce.

Given these gaps, this study introduced *Lakbay Bayani*, a 2D platformer game that integrated AR technology to bring the models of Filipino heroes to life. It featured interactive gameplay and augmented scenes such as 3D hero models that could be accessed by pressing buttons within the game.

This study aimed to evaluate how *Lakbay Bayani* deepened the comprehension of everyone, especially students, regarding Filipino heroism, and encouraged learning through a combination of interactive gameplay and immersive AR features. By focusing on the Philippine educational context, this study provided innovative insights into how local heritage and digital technology came together to make history more engaging, accessible, and meaningful for anyone.

1.2 Purpose and Project Description

The purpose of this study was to design, develop, and evaluate *Lakbay Bayani*, a mobile-based 2D platformer game that integrates Augmented Reality (AR) to promote awareness of Philippine heroism through interactive gameplay. The project aimed to provide an engaging alternative to traditional, text-heavy instructional methods by creating a game that features hero-themed levels, structured difficulty progression, and AR-based visualizations. The system was developed using the Unity engine and C#, with an SQLite database managing player progress, scores, and unlocks. Players progressed through stages inspired by selected national heroes, and completing levels allowed them to access AR-based 3D hero models that served as visual extensions of the game. The study evaluated the system based on the ISO/IEC 25010 software quality model, specifically assessing functional suitability, performance efficiency, reliability, usability,

and security. Through this approach, the study determined whether the developed game met the expected standards of a functional, stable, and user-friendly application.

1.3 Objectives of the Study

This study aimed to design, develop, and implement *Lakbay Bayani*, a mobile-based 2D platformer game that promotes Philippine heroism through game-based learning and augmented reality (AR) for general audiences.

Specific Objectives:

1. To create ten (10) playable levels per hero, divided into three (3) easy, three (3) medium, and four (4) hard stages, ensuring increasing complexity and cognitive engagement.
2. To assign each hero a specific gameplay genre such as action, puzzle-solving and narrative, and strategy that reflects their historical persona, and roles.
3. To include narrative cutscenes at the beginning of each level to provide historical context, character background, and connection to events.
4. To develop a feature that displays a historical object or artifact related to the hero after each level is completed, along with its image and a brief description.
5. To design a hero and item information system that unlocks detailed profiles of national heroes along with encountered artifacts during gameplay in each level, providing historical context and significance upon level completion.
6. To implement a reward system that allows players to collect coins during gameplay and after completing game levels, which can be used to unlock at least one additional hero depending on the number of coins collected.

7. To establish a hero unlocking system that allows players to unlock the next hero upon reaching level seven (7) with the previous hero, or alternatively purchase the hero using in-game coins.
8. To integrate at least one AR interactive scene per hero, allowing players to view a 3D representation of the Filipino hero through their mobile devices.

User Evaluation (Based on ISO 25010 Standards)

User evaluation was conducted using the ISO/IEC 25010 software quality model, focusing on the attributes most relevant to the nature and objectives of Lakbay Bayani. The evaluation emphasized functional suitability, reliability, usability, efficiency, and security, as these directly affected the game's educational effectiveness and player experience.

- **Functional Suitability** – measured by the accuracy and completeness of game features, such as hero unlocking, AR integration, and progress tracking. Respondents rated whether the system met its intended purpose of promoting historical awareness.
- **Reliability** – assessed through system stability, error frequency, and recovery from failures during gameplay. Indicators included smooth transitions between levels and consistent AR activation.
- **Usability** – evaluated by ease of navigation, clarity of instructions, and overall player satisfaction. Metrics included questionnaire ratings on intuitiveness and accessibility for different age groups.
- **Efficiency** – measured by system responsiveness, loading times, and resource utilization during gameplay and AR interactions.

- Security – assessed by the protection of user data stored in the SQLite database, particularly progress and scores.

User evaluation was conducted through a structured questionnaire using a 5-point Likert scale. Respondents included students, educators, and general gamers who participated in playtesting sessions and completed surveys to provide feedback on functionality, usability, and overall performance. This process ensured that the mobile-based educational game prototype was technically sound and effective in promoting engagement, cultural pride, and learning about Philippine heroism.

1.4 Significance of the Study

This study introduced *Lakbay Bayani*, a 2D platformer game that helped players learn about Philippine heroes through interactive and engaging gameplay. Since traditional history learning often relied on passive methods such as reading textbooks or watching documentaries, the game provided a new way to experience the journeys of heroes by overcoming challenges, completing missions, and exploring important historical events. The study was designed to benefit the following:

Players – These included students, educators, and casual gamers of all ages. Players enjoyed a fun gaming experience while learning about national heroes, their sacrifices, and their roles in shaping the nation. Through gameplay, they gained a deeper understanding of heroism and Philippine history in a more memorable and interactive way.

Game Developers and Educational Content Creators – The project served as a model for combining gaming and education effectively. It demonstrated how

games can be used to teach lessons, share values, and present historical narratives in creative and engaging formats.

Future Researchers – They may use this study as a reference for developing similar game-based or AR-enhanced applications. The documentation, system design, and evaluation results can serve as a basis for improving gameplay mechanics, expanding AR functionalities, or exploring additional educational and cultural themes.

1.5 Scope and Limitations

This study focused on the design, development, and implementation of Lakbay Bayani, a mobile-based 2D platformer game that promoted Philippine heroism through game-based learning and augmented reality (AR). The project specifically covered the following scope:

1. The game was designed for general audiences, including learners and individuals interested in exploring and understanding Philippine history and heroism.
2. The game featured unique gameplay mechanics and genre for each selected Filipino national hero, based on their historical roles, traits, and legacy. These mechanics utilized varied formats such as action, puzzle and narrative, and strategy.

1. Lapu-Lapu

- Genre: Action
- Game Type: Melee Combat
- Difficulty: Easy to Hard

2. Jose Rizal

- Genre: Puzzle and Narrative
- Game Type: Word/Image Puzzle
- Difficulty: Easy to Hard

3. Andres Bonifacio

- Genre: Action
- Game Type: Top-Down Tactical Shooter
- Difficulty: Easy to Hard

4. Emilio Aguinaldo

- Genre: Strategy
- Game Type: Tower Defense
- Difficulty: Easy to Hard

5. Apolinario Mabini

- Genre: Puzzle and Narrative
- Game Type: Word Search Puzzle
- Difficulty: Easy to Hard

3. Augmented Reality (AR) was integrated into the game by allowing players to press a button to view a 3D model of each hero using their mobile device.

4. The game will not use pixelated or retro-style sprites. Instead, it will feature clean, modern 2D visuals to better align with the overall educational and immersive goals of the project.

However, it is important to note the limitations of the study:

1. The game only covered a portion of Philippine history, focusing on the Pre-Colonial to early 20th-century periods. Later historical periods are not included in the current version.
2. The game was limited to five (5) Filipino heroes, selected based on their historical significance and suitability for game-based storytelling.
3. The game supported only single-player offline gameplay and did not include multiplayer or online features.
4. The game was developed exclusively for Android devices to ensure accessibility and convenience for mobile users, which limited accessibility for users of other operating systems, such as iOS or desktop platforms.
5. Android devices required a minimum of Android 11 operating system, 4 GB of RAM, and 64 GB of storage to run the game smoothly.
6. AR features were limited to displaying 3D representations of heroes only and did not include full AR environments or landmarks.

1.6 Conceptual Framework

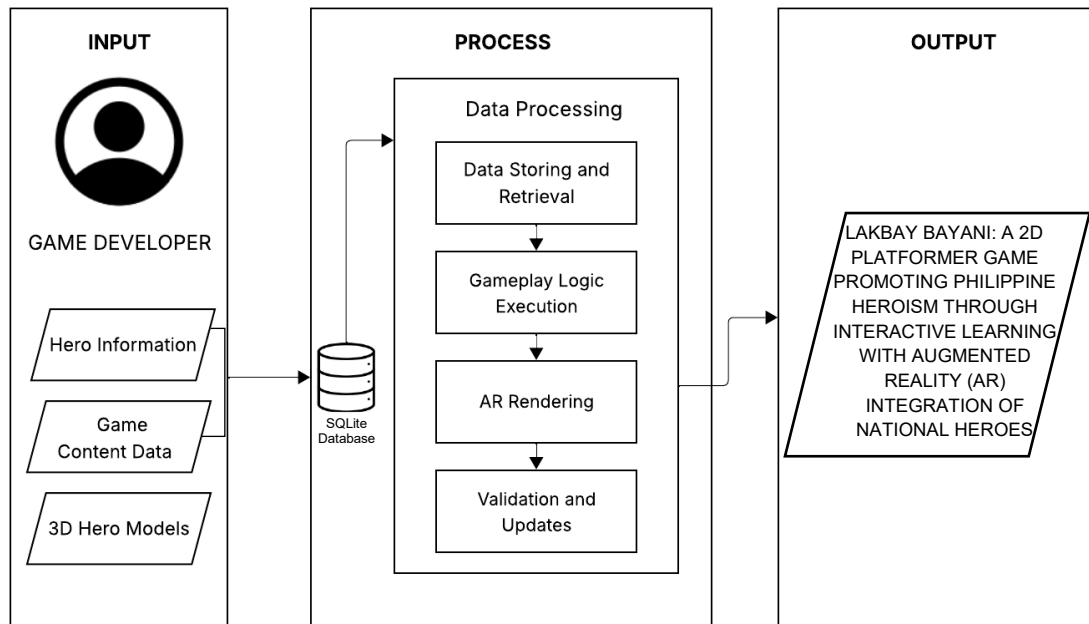


Figure 1.1 Conceptual Framework of the Study

Figure 1.1 shows the conceptual framework of Lakbay Bayani followed the Input–Process–Output (IPO) model to illustrate how developer-provided content was transformed into an interactive educational game. The input included hero information, game content data, and 3D hero models, all of which were stored in a local SQLite database. The process involved retrieving this data, executing gameplay logic using Unity, rendering augmented reality visuals through built-in AR Foundation, and validating player progress throughout the game. These steps worked together to ensure smooth gameplay and immersive learning experiences. The output was a mobile-based 2D platformer game that allowed players to engage with different game genres based on each hero, access historical information, and view 3D hero models through AR.

1.7 Definition of Terms

For the purpose of this study, the following terms are defined to guide the reader's understanding:

2D Platformer	In Lakbay Bayani, it is the game format in which the players utilize a hero avatar to pass through the different stages that are the representation of the significant historical events of the Philippines.
Augmented Reality (AR)	In this project, AR is the technology that shows the 3-dimensional models of Filipino heroes to the users through their mobile devices so that the users can recognize the heroes more easily and get familiar with them.
Game-based Learning	One such educational method is game-based learning which employs game mechanics as a means of learning. The case of Lakbay Bayani is the incorporation of history lessons into missions, quizzes, and challenges which is the essence of the game-based learning approach. Pupils become active participants of the content thus gaining not only more comprehensive but also more enjoyable Filipino history learning.
Gamification	This project utilizes this technique to make history lessons more interesting by allowing players to get

points, level up, and move through historical stages thus learning is facilitated through the interactive play.

Philippine Heroism

This project utilizes this technique to make history lessons more interesting by allowing players to get points, level up, and move through historical stages thus learning is facilitated through the interactive play.

Unity Engine

It is the primary software behind the various stages of Lakbay Bayani, ranging from the conception, the fine-tuning to the final release of the game on different platforms, like mobile devices (Android). Thus, it is the perfect set of tools for the implementation of both the game's mechanics and the user interface.

CHAPTER II

REVIEWOF RELATED LITERATURE

2.1 Literature Map



Figure 2.1 Literature Map

Figure 2.1 above was organized into five major themes to present how previous research supported the development of Lakbay Bayani. First, research on Game-Based Learning (GBL) revealed that games increased student motivation, engagement, and

understanding of lessons. Second, the body of research on Technology Integration in History Education suggested that the use of digital tools made it easier for students to visualize historical events, thereby improving their understanding and interest. Third, investigations on 2D Platformer Games uncovered that this genre became popular in the education sector due to its ease of use, logical progression, and strong potential for integrating educational tasks. Fourth, studies on Philippine Heroism in Learning reached a consensus about the significant role of raising national identity, historical awareness, and cultural pride among learners. Finally, study materials on the use of Augmented Reality (AR) in Learning revealed that AR facilitated immersion, interactivity, and memory retention.

2.2 Review of Related Literature

Game Based Learning in Education

Game-based learning was a method of teaching that involved using games to impart knowledge to students. The many studies have found that this method greatly increases students' interest and their active participation during class. Tamosevicius (2022), for instance, elaborated on the various ways in which game-based learning could be introduced in school and the subsequent advantages of that in the present-day children's education. She emphasized that the method was adaptable, helped the learners to acquire digital skills, and was an excellent medium for students to stay engaged, think critically, and even remember the lessons for a longer period of time. Likewise, Alotaibi (2024) carried out a systematic review and meta-analysis on the topic of game-based learning in early childhood education and his findings showed that such a pedagogical approach

had a positive impact of moderate to large significance on cognitive, social, emotional, and motivational aspects of children's development, thus providing strong evidence for its use in the education sector.

Alduwairaj (2024) noted that while digital game-based learning (DGBL) offers several advantages, some researchers have questioned its effectiveness when compared to traditional teaching methods. Nadeem et al. (2023) investigated the effect of digital game-based learning on student engagement and motivation. They found out that the students' participation in game-based activities was a lot more successful than in traditional instruction, resulting in a greater number of tasks being carried out and students becoming intrinsically motivated to a higher degree as the main reasons why these activities are preferred were the attention capture and the motivation. Similarly, Wang et al. (2022) research involving digital educational game use among primary school students in the STEM subjects revealed an increase in the academic performance of students in science and math. Students participating in the use of educational games scored consistently better than those in traditional learning settings, thus, demonstrating the great potential of digital tools for the advancement of education further.

Game-based learning was thoroughly backed by a wide range of research and has been used across a different set of subjects, such as math, science, and language, primarily in early education. There were a lot of studies designed to research elementary learners since they react well to interactive and creative learning tools like digital games. Such learning tools help engage them and also make it easier for them to comprehend the lesson. Though the majority of research

has been concentrated on the younger students, the advantages of game-based learning are not limited to them only but broadly applicable to all audiences.

Integration of History and Technology in Learning

The integration of technology in history education had gained significant attention in recent years, with numerous studies exploring its potential to improve both teaching and learning. Technology offered various ways to make history more engaging for students. For example, Padli and Yasin (2023) highlighted methods like Outdoor Classroom Learning (OCL), Self-Access Learning (SAL), and the use of Information and Communication Technology (ICT) to enhance history lessons. Similarly, Lee (2023), in the *Muallim Journal of Social Science and Humanities*, discussed how digital tools such as online databases, virtual tours, interactive maps, and multimedia resources provided students with primary and secondary sources, promoting curiosity and active learning.

However, effective integration required teachers to overcome various obstacles. Haron et al. (2021) identified barriers such as limited internet access, inadequate digital infrastructure, and insufficient training for using online platforms. These issues highlighted the need for strong support systems and training programs to help educators adapt to technological changes.

Padli and Yasin (2023) emphasized that in primary schools, history education aimed to build a deeper understanding of identity, values, and historical events. When theory, reading, and activities were combined with interactive tools, students were more likely to engage meaningfully with historical content.

Technology not only enhanced this learning experience but also made history more relatable for younger generations.

Nonetheless, challenges remained. As noted by Padli and Yasin (2023), educators encountered resistance to change and required institutional support to implement digital methods effectively. The Technology in Education – 2023 GEM Report (2023) also drew attention to the global digital divide, which limited access to technology and hindered digital learning. The organization emphasized the need for equitable access and professional development for teachers to build their digital skills.

2D Platformer Games and Their Potential in Education

Game-based learning (GBL) had shown positive effects in education, improving cognitive, social, emotional, and motivational outcomes (Alotaibi, 2024). Research in mathematics education also highlighted that digital games increased students' motivation and engagement (Hui & Mahmud, 2023). Moreover, Earichappan and Hashim (2024) emphasized that GBL fostered 21st-century skills such as collaboration, critical thinking, and digital literacy.

Among game types, 2D platformer games offered unique advantages for education due to their simple yet engaging gameplay, which created immersive and interactive learning environments. This format was especially effective for younger learners, providing enjoyable ways to actively engage with content. Pratama et al. (2022) discussed the creation of a 2D platformer game for Indonesian language instruction, showing how this game type could effectively support learning in different subjects.

Research indicated that game-based learning, including 2D platformers, boosted student motivation and engagement while addressing challenges in traditional education. Albano, Arzarello, and Dello Iacono (2020) highlighted that game-based learning made education more engaging and helped combat disinterest in the classroom. Similarly, Hidayat et al. (2024) supported these findings, noting that game-based learning improved motivation and fostered a more engaging learning environment.

However, challenges remained; poorly designed games could cause negative effects such as addiction or low self-esteem, highlighting the need for thoughtful design to ensure positive learning outcomes. These findings supported the use of 2D platformer games like *Lakbay Bayani* to promote Philippine heroism by combining engaging gameplay with educational content, fostering motivation, historical awareness, and essential skills.

Philippine Heroism and Values

In the Philippines, many people had largely forgotten or were unaware of the rich history and significant contributions of the country's heroes. This fading collective memory had been a concern, especially because these heroes embodied values that remained vital for national identity and social unity.

Several scholars had studied the many sides of heroism in the Philippine context. For example, the Ayala Museum, through its exhibition *Splendor: Juan Luna, Painter as Hero*, showed how public memory of heroes like Juan Luna was shaped by nationalist stories (Magazine, 2023). This suggested that heroism was not fixed but changed over time through social and political views. Similarly,

Andres (2024), in a report by the Friedrich Naumann Foundation, explored how the younger generation saw heroism. The study found that while traditional views focused on sacrifice, there was also growing respect for everyday heroes, those who inspired, brought change, and showed responsibility in daily life.

The National Historical Commission of the Philippines (NHCP), as discussed by Montemayor (2024), confirmed the criteria for recognizing a hero. These included sacrifices for the nation's welfare, the reasons and methods used, moral character, and the lasting impact on society. Though these criteria were developed some time ago, they still helped in understanding historical heroism.

The idea of heroism was also reconsidered after the COVID-19 pandemic. Admin (2024) highlighted how stories about Filipino heroes changed with society's values, especially during and after the pandemic. This showed that heroism was a flexible idea, shaped by current events and public opinion.

Integration of Augmented Reality (AR) in Learning

Augmented Reality (AR) enhanced learning by overlaying digital information onto the real world, creating interactive and immersive experiences. In education, AR was shown to improve student engagement, understanding, and retention of information (Sinha, 2023).

A study by Riego and Morales (2024) at Guinayangan National High School in Quezon found that Grade 7 students who used AR-based learning aids in Biology performed better in posttests compared to those who received traditional instruction. This suggested that AR could be an effective tool in

enhancing students' learning outcomes. In the context of Philippine education, a study by Mantikayan et al. (2022) assessed the effectiveness of an AR-based mobile application in the Bangsamoro Autonomous Region in Muslim Mindanao. The research indicated that the application positively influenced students' attitudes and satisfaction, highlighting the potential of AR in improving educational experiences in the region.

Furthermore, a review by Dhaas (2024) emphasized that AR transformed education by making learning more active and meaningful. The study found that AR applications across various subjects, including history, bridged the gap between traditional teaching methods and modern educational needs.

2.3 Synthesis

The reviewed studies collectively show that game-based learning (GBL), 2D platformer games, and augmented reality (AR) are powerful tools for improving student engagement, motivation, and learning outcomes. GBL is a great way to transform lessons into more interactive and fun sessions, at the same time, it supports the development of students' critical thinking and collaboration skills. 2D platformer games, by virtue of their straightforward yet fascinating design, can easily open up ways for learners to get actively involved in the learning process. At the same time, AR enhances learning by creating immersive experiences that help students visualize and better understand complex ideas.

The literatures pinpoint some major issues that have to be addressed. A majority of research works on GBL and platformer games have been conducted in the fields of math, science, or language, and very few have been done in the area of history education.

There are only a few educational games that weave the local culture and historical background into their content, thus the Filipino heritage is not sufficiently represented in the digital learning tools. AR, although has demonstrated great potentials in education, is hardly used in classrooms in the Philippines particularly in the teaching of national history and heroism.

Moreover, an argument is put forward by the scholars that there is a gradual decrease in the knowledge of the heroes of the Philippines, and this in turn demands the introduction of novel strategies in order to establish the connection of the learners with their cultural identity again. These voids have pointed out the necessity of having Lakbay Bayani projects that not only entertain by using gameplay, AR features, and culturally relevant content but also educate history in a more personal and uplifting way to Filipino students.

2.4 Related Systems

Bayani: Fighting Game also known as Bayani: Kanino Ka Kakampi?



Figure 2.2. Sample Screen of Bayani: Kanino Ka Kakampi?
(Wejebe, 2019)

Bayani: Fighting Game, developed by Ranida Games, is a 2D anime-style combat game featuring futuristic versions of Philippine heroes like Jose Rizal and Andres Bonifacio. Set in an alternate timeline, the game focuses on action-packed battles and stylized storytelling to promote cultural identity and national pride (Vivas, 2020). However, it is not designed to educate players about historical facts. In contrast, Lakbay Bayani aims to let players aware and learn especially children and students about real historical events through mission-based gameplay. While both games showcase Philippine heroes, Lakbay Bayani emphasizes accurate historical content and educational value, making it more suitable for learners rather than purely for entertainment.

An Educational Game for Android about the Philippine History Focusing on the Timeline of Jose Rizal and Lapulapu.



Figure 2.3 Sample Screen of Educational Game about the Philippine History Focusing on the Timeline of Jose Rizal and Lapulapu (Jitsuuuuu7, 2022)

This Android-based educational game centers on important events in Philippine history, particularly the lives of two national heroes, Jose Rizal and

Lapulapu (Jitsuuuuu7, 2022). Developed by IT students, it aims to make learning history more interactive by combining storytelling, simple gameplay, and visual aids. The game was intended for wide access through the Google Play Store, supporting gamified learning trends that engage young users through mobile devices.

Compared to Lakbay Bayani, which includes multiple heroes and diverse historical periods transformed into platforming levels with missions and puzzles, this game focuses narrowly on a historical timeline with limited interactivity. While Lakbay Bayani offers a more immersive and action-based experience by involving players in the heroes' journeys, the timeline game acts more like a digital textbook, using guided storytelling to present information. Thus, Lakbay Bayani emphasizes active learning through gameplay, while this project leans more toward informational presentation.

Filipino mobile game 'Galà' teaches players about Philippine history and culture

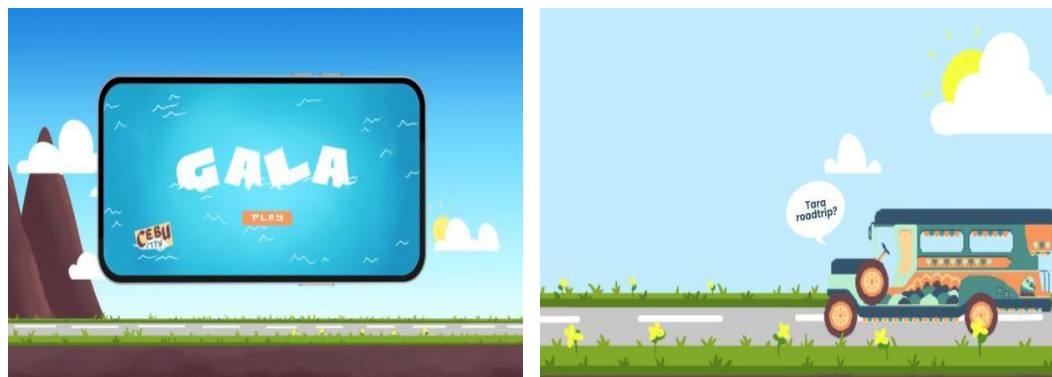


Figure 2.4 Sample Screen of Galà Game (KendiKorp, 2022)

Galà is a Filipino mobile game developed by KendiKorp in partnership with the Cultural Center of the Philippines. The game offers an interactive cultural road trip where players visit 17 iconic landmarks across the Philippines, all presented in a pixel-art style. Players fuel a magical jeepney with milk tea ingredients, adding a fun twist as they learn about regional myths, historical artifacts, Filipino culture, and the traditional jeepney design. Available for offline play and through the Google Play Store, Galà is accessible to many users. While both Galà and Lakbay Bayani promote Filipino heritage, their approaches differ: Galà focuses on cultural exploration and leisurely learning about regional traditions and landmarks, whereas Lakbay Bayani offers mission-based gameplay centered on national heroes and important historical events. Lakbay Bayani encourages active problem-solving and engagement with the values and sacrifices of the heroes, making it more action-oriented and historically immersive, while Galà provides a relaxed, exploratory educational experience highlighting the country's diverse cultures and history.

Monument Valley Game



Figure 2.5 Sample Screen of Monument Valley Game (Bellingham, 2022)

Monument Valley is a puzzle game where players help a character move through strange and beautiful paths by shifting parts of the level. The game teaches problem-solving and thinking from different angles. For Lakkay Bayani, this gameplay can be applied by creating puzzle levels where players unlock parts of a hero's story or complete tasks like finding secret documents, passing through mazes, or solving symbol-based puzzles related to Philippine culture and history.

Sample Screen of Pokémon GO Game



Figure 2.6 Sample Screen of Pokémon GO (Freese, 2021)

Pokémon GO is a popular AR mobile game where players explore real-world locations to find and catch virtual Pokémon, combining location-based gameplay with augmented reality visuals for an immersive experience. In Lakkay Bayani, this idea is adapted by using AR features to display 3D representations of Philippine heroes when players interact with certain game elements. However, the AR functionality is limited to showing these 3D hero models and does not include full AR environments or real-world landmarks.

Table 2.1 Comparison Matrix of Related Systems

RRS	Educational Focus	Philippine Heroes	Game Genre	Augmented Reality	Reward System
Bayani: Kanino Ka Kakampi?	Moderate	Yes	Fighting	None	Unlocks
Timeline of Rizal & Lapu-Lapu	High	Yes	Interactive Timeline	None	None
Galà'	Low	No	Cultural Exploration	Yes	Exploration-based rewards
Pokémon GO	Low	No	AR Exploration	Yes	Collectibles
Lakbay Bayani	High	Yes	2D Platformer + AR	Yes	Coins + Hero Unlocks

The systems compared illustrate diverse approaches to integrating Philippine history and culture into digital games. Bayani: Kanino Ka Kakampi? emphasizes stylized combat and hero representation, while Timeline of Rizal & Lapu-Lapu focuses on chronological learning through interactive timelines. Galà' highlights cultural exploration using AR, encouraging experiential engagement with heritage, whereas Pokémon GO popularized AR exploration globally but without a direct educational or cultural focus.

Among these, Lakbay Bayani distinguishes itself by combining high educational value with mission-based gameplay centered on Philippine heroes. Its integration of interactive levels, puzzles, and augmented reality creates a dynamic learning environment where historical narratives are not only presented but actively experienced. The inclusion of a reward system through coins and hero unlocks further motivates learners, reinforcing engagement while deepening cultural understanding. This unique blend of education, interactivity, and AR immersion positions Lakbay Bayani as a highly effective tool for both learning and entertainment.

CHAPTER III

REVIEWOF RELATED LITERATURE

3.1 Software Methodology

Considering the time frame allotted for the proponents to complete this project as well as the nature of the project, the proponents decided to follow the Agile SDLC, implemented through the Scrum methodology. Scrum was chosen due to its iterative and flexible structure, which allowed for continuous feedback, rapid development, and adaptive planning, which was ideal for a game-based educational application like Lakbay Bayani. The Scrum process was a popular Agile method that allowed teams to focus on continuous improvement while building and iterating quickly (Martins, 2025).

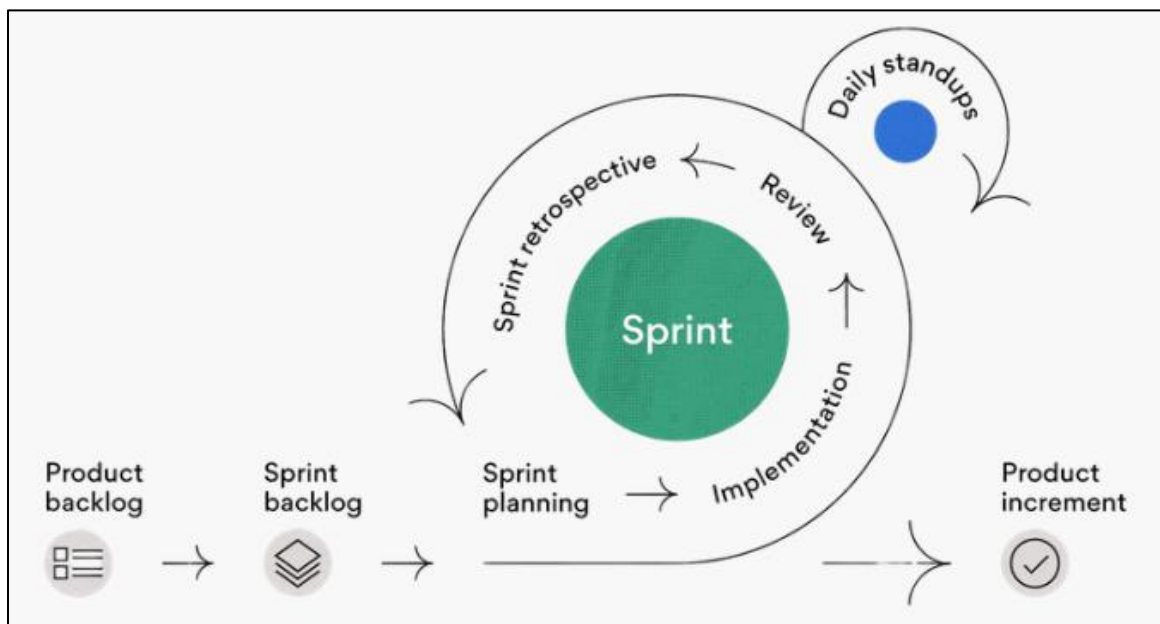


Figure 3.1: Scrum framework (Martins, 2025)

Scrum broke down the development process into small units called sprints, which usually lasted from one to two weeks. A sprint was started with sprint planning, where the

team members decided which tasks from the product backlog a list of features and requirements prioritized by value should be done. These selected tasks became the sprint backlog, which was like a team's work plan during the sprint. Teams met daily at stand-up meetings to check progress, solve problems, and change plans if necessary. The review and retrospective were the last events of the sprint, assembled to evaluate the work done and to plan better future cycles. The outcome of every sprint was a product increment, a working version of the game which included new features or improvements.

3.2 System Planning

The project was conceptualized in response to the growing need for more engaging, technology-enhanced tools to teach Philippine history, as traditional methods such as textbook reading and lectures often failed to capture student interest. Inspired by the potential of game-based learning and AR technology, the proponents envisioned a mobile application that would combine interactive gameplay with immersive visualizations of Filipino heroes, thereby fostering cultural pride and deeper historical understanding among learners.

The preparation stage, the team members pinpointed the main features that were in line with the educational objectives set by the project. These features included hero-based levels, 2D platforming mechanics, collectible historical facts, and AR interactions enabling players to see three-dimensional representations of national heroes. In order to be consistent and technically feasible, the history-based game was plotted according to the Philippine historical timeline with each hero getting the gameplay mechanics that mirrored their historical roles.

Feasibility Study

The development of Lakbay Bayani: A 2D Platformer Game Promoting Philippine Heroism through Interactive Learning with AR Integration was assessed for feasibility across operational, economic, and technical dimensions to ensure that it could be realistically developed, implemented, and maintained within available resources and constraints.

Operational Feasibility

The target users of the history game application were students, educators, casual players, and game developers interested in interactive learning. The project included an intuitive tutorial system that guided users on how to navigate the game mechanics and augmented reality features. In addition, the application was seamlessly integrated into various usage contexts by allowing students and educators to use it as a supplementary educational tool while enabling casual players to enjoy it as standalone entertainment.

Users were engaged through interactive levels and missions where they learned historical facts and earned rewards. The operation of the application was designed for flexible use, though primary usage correlated with academic schedules for student and educator users, while casual usage remained consistent throughout the year. Users, as a whole, benefited from increased engagement and knowledge retention compared to traditional learning methods, and they had consistent access to an innovative educational platform.

Economic Feasibility

Shown in Table 3.1 is the cost-benefit analysis used to examine the project's feasibility from an economic perspective.

Table 3.1 Cost / Benefit Analysis

COSTS	
CATEGORY	ESTIMATED BUDGET
1. Product Backlog	1,350
2. Sprint Backlog	1,000
3. Sprint Planning	-
4. Implementation	1,300
5. Sprint Review	-
5.1 Expert Review	500
6. Product Retrospective	1,000
7. Daily Standups	-
8. Product Increment	-
Total Cost:	5,150

The economic feasibility of Lakbay Bayani: A 2D Platformer Game Promoting Philippine Heroism through Interactive Learning with Augmented Reality (AR) Integration of National Heroes was evaluated through a comprehensive Cost–Benefit Analysis (CBA) to determine whether the system is financially viable and sustainable for educational use. The analysis considered all relevant development and operational costs, including Product Backlog and Research (₱1,350.00), which covered research activities and the purchase of reference books and academic materials necessary for the study. Additional costs included Sprint Backlog (₱1,000.00), Implementation (₱1,300.00), Sprint Review (₱500.00), and Sprint Retrospective with User Evaluation (₱1,000.00). Other Scrum phases such as Sprint Planning, Daily Standups, and Product Increment were accounted for in the methodology but did not require direct monetary allocation during the initial development cycle. The total

estimated project cost amounted to ₱5,150.00, which was deemed reasonable given the scope of the project and its expected cultural and educational impact.

On the other hand, the anticipated benefits of Lakbay Bayani were not measured in direct monetary profit but in its cultural and educational contributions. The system was envisioned to promote Philippine history and heroism by transforming learning into an interactive and engaging experience through gameplay and augmented reality. It was expected to enhance historical awareness, motivate learners through challenges and rewards, and provide opportunities for integration into schools and tourism initiatives. By making history meaningful, accessible, and immersive, the project aimed to deliver long-term value aligned with its mission to inspire appreciation of national identity and heritage, thereby demonstrating strong social and educational returns on investment.

Technical Feasibility

To ensure smooth development and deployment, Lakbay Bayani adopts technologies that are already proven effective in game-based learning and augmented reality applications. Unity Engine serves as the core development platform, enabling the creation of 2D platformer levels, hero-specific mechanics, narrative cutscenes, and collision-based interactions. For the augmented reality features, the built-in AR Foundation framework is utilized to detect surfaces or markers and to render 3D hero visualizations that enhance storytelling and historical immersion.

The AR modules are specifically designed to run on AR Core-supported Android devices, ensuring compatibility with smartphones that meet the minimum requirements of Android 11, 4 GB RAM, and 64 GB storage. This dependency guarantees that augmented reality functions such as hero visualization and artifact interaction operate smoothly on capable devices. Meanwhile, the main 2D gameplay is optimized for Android platforms in general, with performance scaling depending on the device version. Although AR features are limited to AR Core-enabled phones, the core platformer mechanics remain accessible to a wider range of Android devices, thereby balancing inclusivity with advanced functionality.

Work Breakdown Structure (WBS)

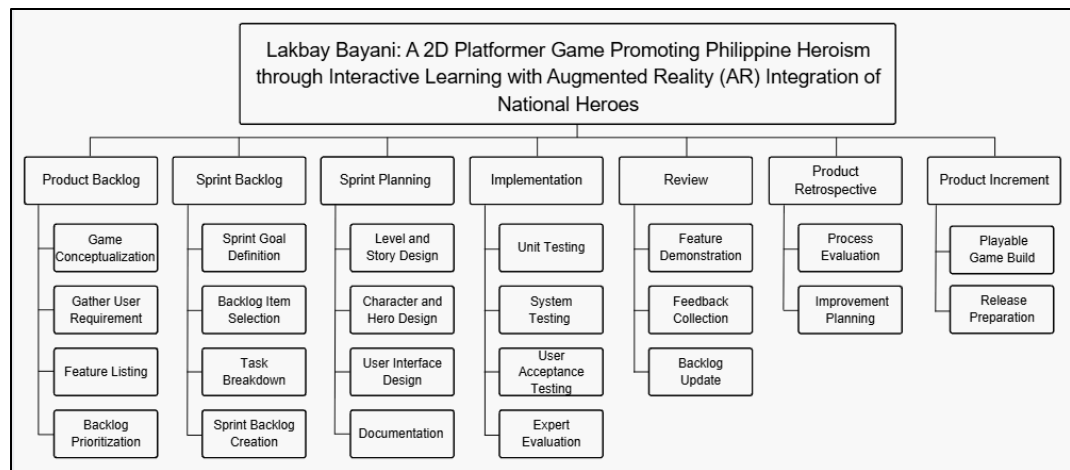


Figure 3.2: Work Breakdown Structure

Figure 3.2 presents the Work Breakdown Structure of Lakbay Bayani: A 2D Platformer Game Promoting Philippine Heroism through Interactive Learning with Augmented Reality (AR) Integration of National Heroes, developed using the

Scrum methodology. The project is organized into eight major Scrum components: Product Backlog, Sprint Backlog, Sprint Planning, Implementation, Review, Product Retrospective, Daily Standups, and Product Increment.

The development process begins with the Product Backlog, where the proponents conducted game conceptualization, gathered user requirements, listed key features, and prioritized backlog items based on importance and feasibility. These activities established the foundation for iterative development. The Sprint Backlog was then created by defining sprint goals, selecting backlog items, breaking them down into tasks, and finalizing the sprint backlog for implementation.

During Sprint Planning, the team focused on level and story design, character and hero design, user interface design, and documentation to prepare the selected backlog items for development. The Implementation phase involved unit testing, system testing, user acceptance testing, and expert evaluation, ensuring that each feature met functional, educational, and usability requirements within the sprint cycle.

The Review phase included final setup, distribution, and post-deployment monitoring, where completed features were demonstrated and evaluated to assess system readiness and performance. Following this, the Product Retrospective was conducted to evaluate the development process and identify areas for system improvement and planning enhancements in future sprints. Daily Standups were held throughout the sprint cycles to evaluate progress, address issues, and plan improvements collaboratively.

Each sprint concluded with a Product Increment, resulting in a playable game build and release preparation. Through continuous iterations, the Scrum-based Work Breakdown Structure enabled flexible development, timely feedback

integration, and the progressive enhancement of Lakbay Bayani as a functional and educational game system.

Gantt Chart

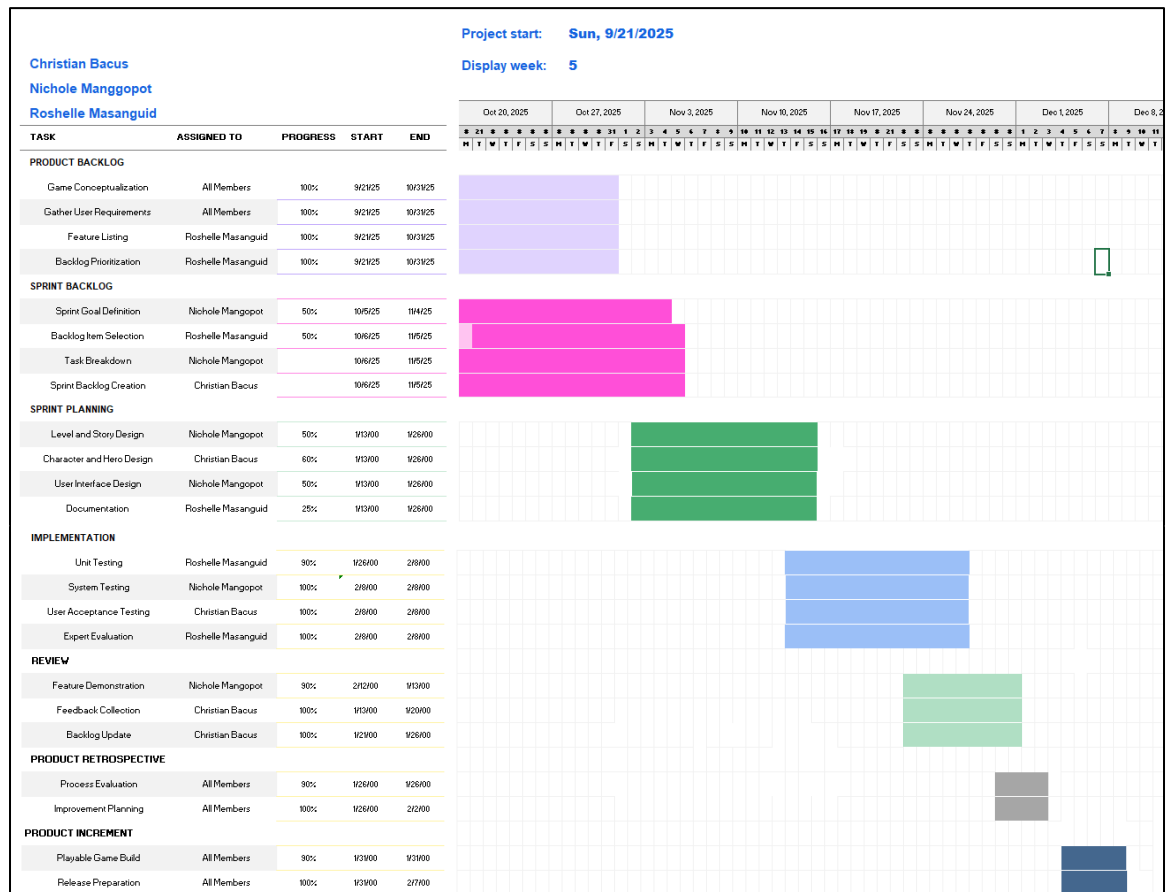


Figure 3.3 Gantt Chart

Figure 3.3 was structured into six key phases: Research Planning, Design, System Development, Testing, Deployment, and Maintenance. These phases were distributed across multiple sprints following Scrum methodology, allowing the team to deliver functional increments at the end of each cycle. The Research Planning phase established the foundation through stakeholder identification, requirement gathering, feasibility analysis, and project planning. Design focused on game conceptualization, level and story content, UI design, and technical architecture. System Development involved asset

creation, coding, integration, and documentation. Testing ensured system reliability through unit tests, user acceptance, and expert evaluation. Deployment covered final release and post-deployment monitoring, while Maintenance addressed ongoing updates and user feedback integration. This structured flow ensured that each phase contributed to the progressive achievement of project objectives within the academic timeline.

3.3 System Analysis

The system analysis phase focused on examining the existing game to understand how it functioned and to identify areas that need improvement.

Description of the Current System

The existing system for learning Philippine heroes primarily relied on manual and traditional instructional methods commonly used in classroom settings. The learning process typically began with teachers introducing heroes through textbooks, printed modules, lectures, and visual aids such as posters or charts. Students acquired information by reading assigned materials, listening to discussions, and memorizing facts such as the hero's biography, historical contributions, and significant events. Assessment of learning was conducted through written quizzes, recitations, and paper-based examinations.

While this system effectively delivered factual information, several limitations affected student engagement and learning retention. First, the learning process was largely teacher-centered, offering limited opportunities for active participation and experiential learning. Second, historical lessons were often presented in a static and linear manner, making it difficult for students to visualize historical events and fully connect with the experiences of Filipino heroes. Third,

the absence of interactive elements reduced student motivation and curiosity, especially among younger learners who are more inclined toward digital and game-based environments. Lastly, the manual system lacked immersive and exploratory features that could foster deeper historical understanding, emotional connection, and cultural pride.

These limitations highlighted the need for an innovative approach to teaching Philippine heroism. To address these gaps, the proposed system Lakbay Bayani integrates Game-Based Learning (GBL) and Augmented Reality (AR) to transform traditional lessons into interactive missions, story-driven gameplay, and immersive hero experiences. Through this approach, learning moves beyond passive information delivery and becomes an engaging, meaningful, and technology-enhanced educational experience.

Business Process Modelling Notations

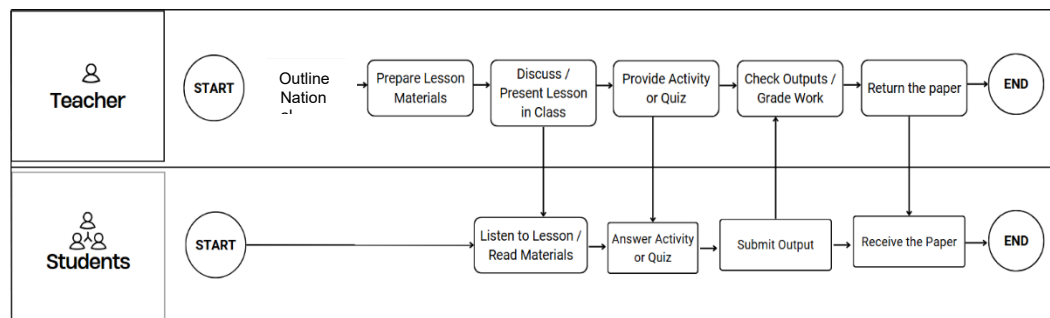


Figure 3.4 Business Process Modelling Notations

Requirement Specification

The proposed system, Lakbay Bayani, was designed to meet both functional and non-functional requirements to ensure that it delivered an

engaging, reliable, and educational gaming experience. These requirements defined what the system must accomplish and the standards it had to meet in order to address the limitations of the current approach.

1. Functional Requirements

The system was expected to perform the following core tasks and operations based on user needs and the intended learning outcomes:

- The system allowed players to select a Filipino national hero and play through levels based on that hero's historical background.
- The system allowed players to move, jump, attack, solve puzzles, or perform actions depending on each hero's assigned gameplay style.
- The system displayed narrative cutscenes before each level to explain key historical events.
- The system unlocked new levels and heroes after players completed required objectives.
- The system stored and retrieved player progress so users could continue playing without starting over.
- The system integrated Augmented Reality (AR) features that enabled players to view a 3D model of each hero using their mobile device.
- The system displayed educational information such as hero biographies, achievements, and historical contributions.

2. Non-functional Requirements

The non-functional requirements of Lakbay Bayani were evaluated based on the ISO/IEC 25010:2011 Product Quality Model, which defines the five core software quality attributes.

Functional Suitability – The game provided all necessary functions to support progression, character interaction, event triggering, and mission completion. Each feature was expected to perform correctly and contribute to smooth gameplay.

Performance Efficiency – The system maintained responsive gameplay with minimal delays, fast loading times, and stable frame rates. Efficient use of memory, processing power, and battery resources was required for consistent performance across supported devices.

Usability – The game offered user-friendly interactions, enabling players to navigate menus, control characters, and understand mechanics easily. Interfaces and visuals were designed for clarity, consistency, and accessibility, particularly for younger audiences.

Reliability – The system was required to run stably, avoiding crashes or glitches, while ensuring that player progress could be saved and restored even after unexpected interruptions.

Security – The game protected user information and stored data from unauthorized access or alterations, ensuring secure handling and integrity of gameplay records.

Proposed Solution

The proposed system addressed the limitations of the existing approach by introducing Lakbay Bayani, an interactive, mobile-based learning platform that combined game-based learning with augmented reality (AR). Rather than replacing passive learning methods such as textbooks, the system served as an alternative tool that offered a more engaging environment where players could explore historical events, interact with Filipino heroes, and complete mission-based challenges that reinforced educational outcomes. It improved efficiency by storing player progress and achievements in a local database for offline access, enhanced accuracy by aligning gameplay mechanics with verified historical narratives, and strengthened usability through intuitive controls and clear visuals.

The overall workflow was streamlined, allowing players to move seamlessly from hero selection to gameplay and AR viewing within a single application. The system provided significant benefits and value to a wide range of users, including students, educators, and casual gamers. Its implementation was both necessary and feasible, given the accessibility of mobile devices and the growing demand for interactive educational technologies. By integrating entertainment with structured learning, Lakbay Bayani offered a scalable and practical solution to declining interest in Philippine history while ensuring inclusivity for diverse audiences.

3.4 System Design

The system design outlined the structure and operation of Lakbay Bayani, showing how its components interact to deliver cohesive gameplay and augmented reality features.

System Architecture

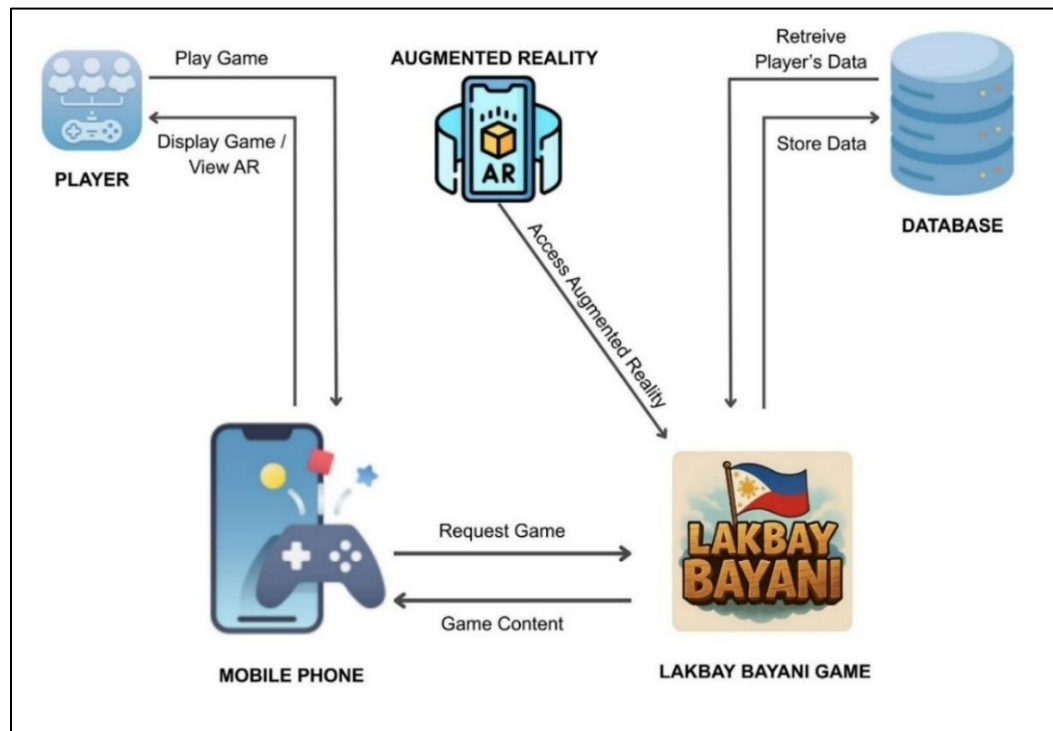


Figure 3.5 System Architecture

Shown in Figure 3.5 was the system architecture of Lakbay Bayani. The system was designed as a mobile-based application, with all core functions processed locally on the player's device. The mobile phone acted as the main client and interface, enabling the player to interact with the game. Game content such as missions, hero information, and educational material was delivered through the Lakbay Bayani Game module, while the Augmented Reality (AR) module, powered by Unity's AR Foundation, rendered 3D hero models and

artifacts using the device's camera. A local SQLite database stored and retrieved player progress, unlocked heroes, and collected items, ensuring that gameplay could continue seamlessly across sessions.

This architecture eliminated the need for constant internet connectivity, allowing smooth gameplay and immersive AR experiences even in offline settings. By keeping all components integrated within the mobile application, the system ensured efficiency through reduced latency, scalability by supporting future content updates, and data security by maintaining player information within the device. The clear separation of the client interface, game logic, AR rendering, and database management provided a cohesive structure that supported both educational and entertainment objectives.

Entity Relationship Diagram

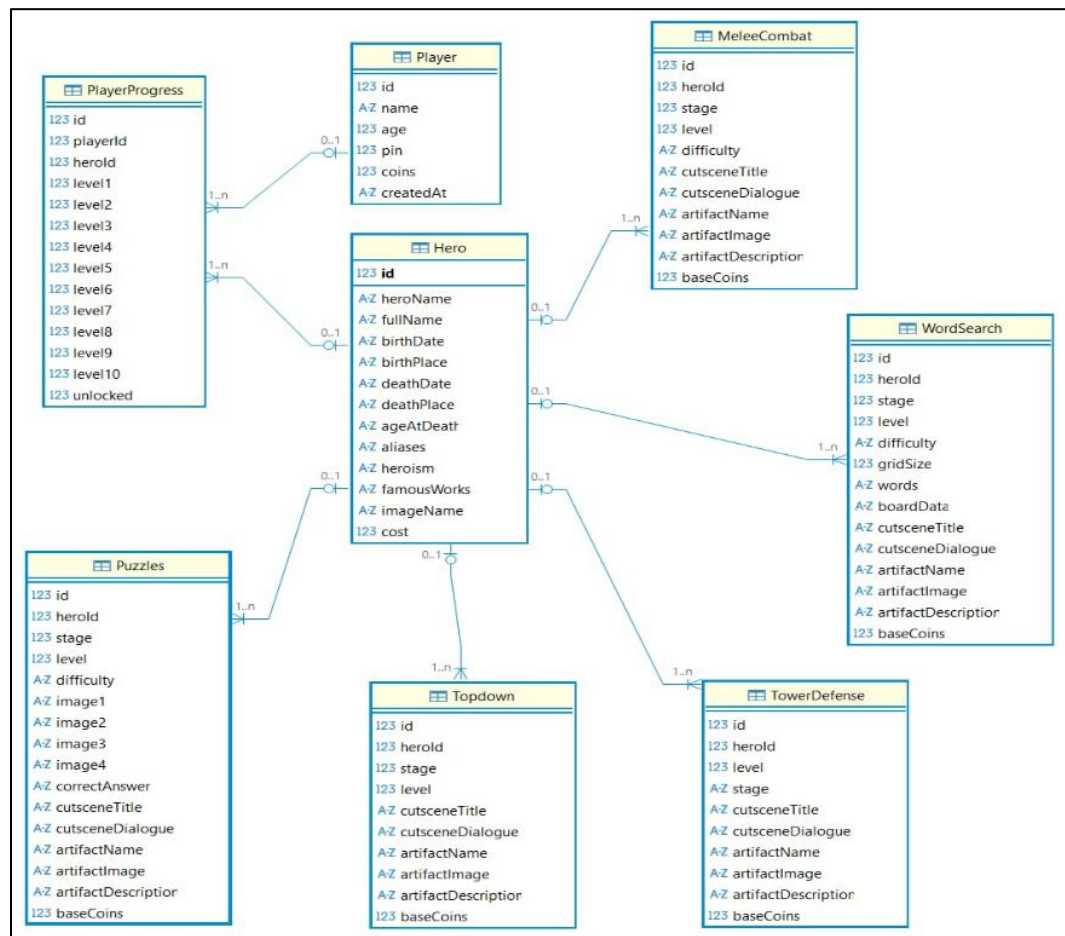


Figure 3.6 Entity Relationship Diagram of Lakbay Bayani

Presented in Figure 3.6 was the Entity Relationship Diagram (ERD) of the Lakbay Bayani game, generated using SQLite and illustrated with Crow's Foot notation. The ERD demonstrated the logical structure of the database, showing how core entities such as Player, Hero, PlayerProgress, and various Game Modes were interconnected through primary and foreign key relationships.

The Player table stored basic user information, while the PlayerProgress table tracked each player's progress across different heroes and levels. The Hero table contained detailed historical data, including birthplace, birth and death dates,

and notable achievements. Each game mode, MeleeCombat, WordSearch, Puzzles, Topdown, and TowerDefense was linked to the Hero table via foreign keys, allowing the system to assign gameplay mechanics based on the hero's historical role and traits.

This relational model supported data consistency by enforcing referential integrity between tables, ensuring that progress and gameplay data were accurately linked to valid player and hero records. It enabled scalability by allowing new heroes and game modes to be added without disrupting the existing schema. The use of SQLite ensured efficiency in data retrieval and storage, particularly for mobile-based applications, as it allowed fast access to player data and game content without requiring a network connection.

Overall, the database design provided a structured and reliable foundation for managing educational content, player interactions, and game progression, aligning with the system's goal of delivering an engaging and historically accurate learning experience.

Data Dictionary

Presented below are the data dictionary tables, which provide detailed descriptions of the data objects and attributes used in the system's data model. The data dictionary serves as a central reference for developers, analysts, and other stakeholders involved in the design and implementation of the game. It ensures a consistent understanding of each field's purpose, data type, format, and relational constraints across all modules.

Table 3.2 Data Dictionary of A 2D Platformer Game Promoting Philippine Heroism through Interactive Learning with Augmented Reality (AR) Integration of National Heroes

TABLE NAME	ATTRIBUTE NAME	CONTENT	TYPE	FORMAT	PK	FK
Player	id	Player's unique ID	int	0	✓	
	name	Player's Name	VARCHAR	xxxxxx		
	age	Player's Age	Int	00		
	pin	Player's unique PIN	VARCHAR	0000		
	coins	Player's collected coins.	int	0000		
	createdAt	Player's logs	DATETIME	dd-mm-yyyy: hh:mm:ss		
Hero	id	Hero's ID	int	0	✓	
	heroName	Hero's Name	VARCHAR	xxxxxx		
	fullName	Hero's Full name	VARCHAR	xxxxxxxxxx x		
	birthDate	Hero's Date of Birth	VARCHAR	mm-dd-yyyy		
	birthPlace	Location where the hero was born	VARCHAR	xxxxxxxxxx		
	deathDate	Hero's Date of Death	VARCHAR	mm-dd-yyyy		
	deathPlace	Location where the hero died	VARCHAR	xxxxxxxxxx		
	ageAtDeath	Hero's age at time of death	VARCHAR	00		
	heroism	Summary of heroic deeds	VARCHAR	xxxxxxxxxx xxxx		
	cost	Costs of each hero	int	0000		
	imageName	Hero's image	VARCHAR	xxxxxx		
Player Progress	id	Player progress ID	int	0	✓	
	playerId	Player's linked ID	int	0		✓
	heroId	Hero's linked ID	int	0		✓
	level1	Progress for level 1	int	0		
	level2	Progress for level 2	int	0		
	level3	Progress for level 3	int	0		
	level4	Progress for level 4	int	0		
	level5	Progress for level 5	int	0		
	level6	Progress for level 6	int	0		
	level7	Progress for level 7	int	0		
	level8	Progress for level 8	int	0		
	level9	Progress for level 9	int	0		
	level10	Progress for level 10	int	0		

Puzzles	id	Puzzle ID	int	0	✓	
	herold	Hero's ID	int	0		✓
	stage	Difficulty indicator	String	Easy/ Medium/ Hard		
	level	Level number	int	00		
	image1	First puzzle image	VARCHAR	xxxxxx		
	image2	Second puzzle image	VARCHAR	xxxxxx		
	image3	Third puzzle image	VARCHAR	xxxxxx		
	image4	Fourth puzzle image	VARCHAR	xxxxxx		
	correctAnswer	Correct word answer	VARCHAR	xxxxxx		
	cutsceneTitle	Title for cutscene	VARCHAR	xxxxxx		
	cutsceneDialogue	Title of the Dialogue	VARCHAR	xxxxxx		
	artifactName	Name of the Artifact	VARCHAR	xxxxxx		
	artifactImage	Image of the Artifact	VARCHAR	xxxxxx		
	artifactDescription	Description of the Artifact	VARCHAR	xxxxxx		
	baseCoins	Collected Coins	int	0000		
Melee Combat	id	Melee Combat ID	int	0	✓	
	herold	ID of the Hero	int	0		✓
	stage	Difficulty indicator	String	Easy/ Medium/ Hard		
	level	Number level	int	00		
	cutsceneTitle	Title of the Cutscene	VARCHAR	xxxxxx		
	cutsceneDialogue	Dialogue in the Cutscene	VARCHAR	xxxxxx		
	artifactName	Name of the Artifact	VARCHAR	xxxxxx		
	artifactImage	Image of the Artifact	VARCHAR	xxxxxx		
	baseCoins	Collected Coins	int	0000		
Word Search	id	Word Search ID	int	0	✓	
	herold	ID of the Hero	int	0		✓
	stage	Difficulty indicator	String	Easy/ Medium/ Hard		
	level	Number level	int	00		
	gridSize	The size of the square grid	int	00		
	words	Words in the Gameplay	VARCHAR	xxxxxx		
	boardData	Representing the entire letter grid of the puzzle	VARCHAR	xxxxxx		
	cutsceneTitle	Title of the Cutscene	VARCHAR	xxxxxx		
	cutsceneDialogue	Title of the Dialogue	VARCHAR	xxxxxx		

	artifactName	Name of the Artifact	VARCHAR	xxxxxx		
	artifactDescription	Description of the Artifact	VARCHAR	xxxxxx		
Topdown	id	Topdown ID	int	0	✓	
	herold	ID of the Hero	int	0		✓
	stage	Stage level	String	Easy/ Medium/ Hard		
	level	Number level	int	00		
	cutsceneTitle	Title of the Cutscene	VARCHAR	xxxxxx		
	cutsceneDialogue	Dialogue in the Cutscene	VARCHAR	xxxxxx		
	artifactName	Name of the Artifact	VARCHAR	xxxxxx		
	artifactImage	Image of the Artifact	VARCHAR	xxxxxx		
	artifactDescription	Description of the Artifact	VARCHAR	xxxxxx		
	baseCoins	Coins collected	int	0000		
Tower Defense	id	Tower Defense ID	int	0	✓	
	herold	ID of the Hero	int	0		✓
	stage	Stage number	String	Easy/ Medium/ Hard		
	level	Number level	int	00		
	cutsceneTitle	Title of the Cutscene	VARCHAR	xxxxxx		
	cutsceneDialogue	Dialogue in the Cutscene	VARCHAR	xxxxxx		
	artifactName	Name of the Artifact	VARCHAR	xxxxxx		
	artifactImage	Image of the Artifact	VARCHAR	xxxxxx		
	artifactDescription	Description of the Artifact	VARCHAR	xxxxxx		
	baseCoins	Coins collected	int	0000		

Data Flow Diagram

The data flow diagram (DFD) provided a conceptual overview of how information moves within Lakbay Bayani. Using the Gane and Sarson methodology, it illustrated the interaction between external entities, internal processes, and the data store, ensuring that user actions such as registration,

gameplay, and progress tracking are processed efficiently and organized into meaningful outputs.

Data Flow Diagram Level 0

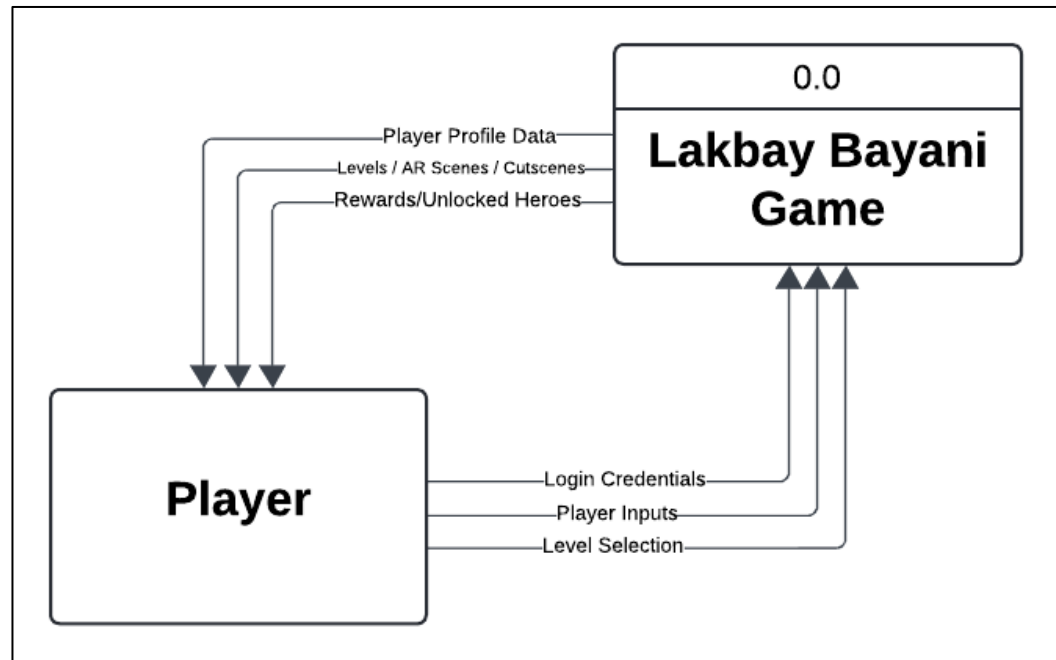


Figure 3.7 Data Flow Diagram Level 0

Illustrated in Figure 3.4 was the Level 0 Data Flow Diagram of Lakbay Bayani, constructed using the Gane and Sarson methodology. This context diagram visualized the system as a single central process, Lakbay Bayani Game, interacting with one external entity, the Player. It depicted the bidirectional flow of information between the player and the system, clarifying how inputs were processed and outputs were delivered.

From the player's side, the system received login credentials, level selections, and gameplay inputs. In response, the game provided player profile data, levels with AR scenes and cutscenes, and rewards such as unlocked heroes

and collected coins. This simplified representation emphasized the core interaction loop and highlighted how the system managed user engagement, content delivery, and progress tracking. The diagram served as a foundational reference for understanding the system's functional boundaries and data exchange mechanisms.

Data Flow Diagram Level 1

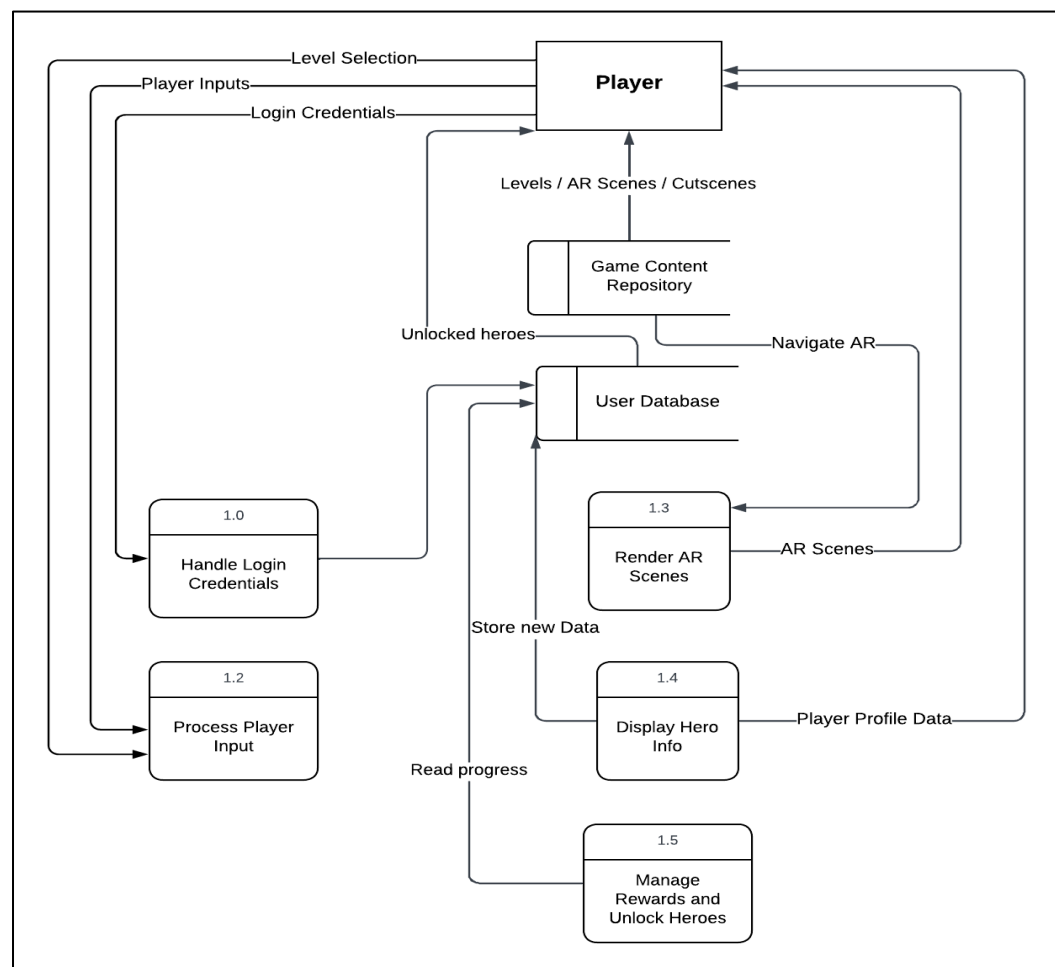


Figure 3.8 Data Flow Diagram Level 1

Illustrated in Figure 3.8 was the Level 1 Data Flow Diagram of Lakbay Bayani, developed using the Gane and Sarson methodology. This diagram

decomposed the system into detailed subprocesses that managed player interactions and internal operations. The external entity, the Player, provided inputs such as login credentials, level selection, and gameplay actions. These were processed by distinct modules including Handle Login Credentials, Process Player Input, Render AR Scenes, Display Hero Info, and Manage Rewards and Unlock Heroes.

Each process communicated with relevant data stores such as the User Database and the Game Content Repository, ensuring that player data, progress, and educational content were accurately retrieved and updated. The flow of information from input to processing to output was designed to maintain data accuracy, secure access, and efficient system performance. By clearly mapping how data moved between users, processes, and repositories, the diagram supported the system's goal of delivering a seamless, interactive, and educational experience.

Use Case Diagram

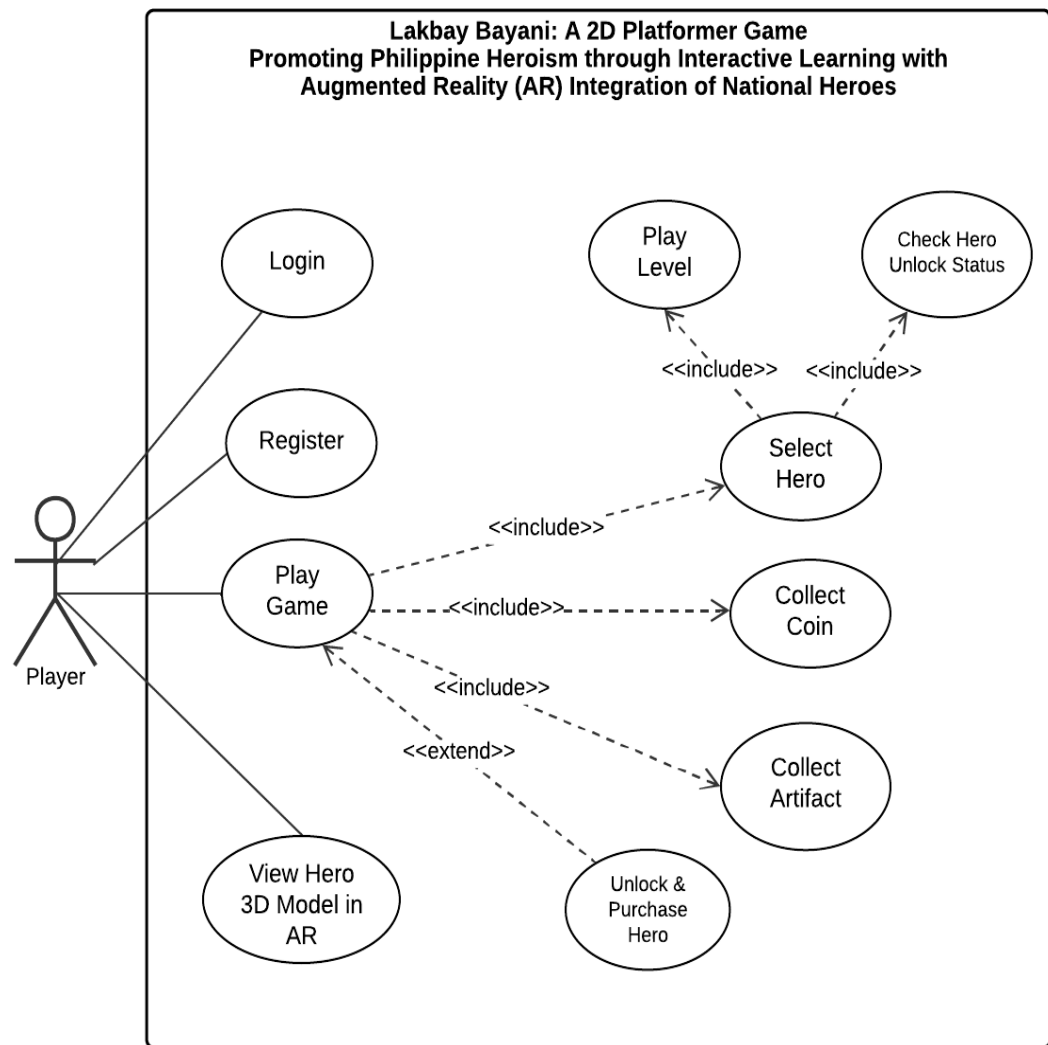


Figure 3.9 Use Case Diagram

Shown in Figure 3.9 was the Use Case Diagram developed for Lakbay Bayani: A 2D Platformer Game Promoting Philippine Heroism through Interactive Learning with Augmented Reality (AR) Integration of National Heroes. The diagram represented the system's functionalities from the user's perspective, using standard notation to illustrate actor interactions, system boundaries, and relationships such as «include» and «extend».

The primary actor identified was the Player, who interacted with the system through several core use cases: Login, Register, Play Game, and View Hero 3D Model in AR. The Play Game use case included sub-functions such as Select Hero, Play Level, and Check Hero Unlock Status, while Collect Artifact extended to Unlock & Purchase Hero. These use cases aligned with the user requirements established during system analysis, ensuring that the player could engage with historical content, complete educational challenges, and access AR features. The diagram effectively visualized how the system supported interactive learning and cultural engagement through structured gameplay.

Use Case Description

In accordance with the use case diagram in the previous section, the proponents provided descriptions for each use case in a form of a table which is shown below.

Table 3.3 Tabular Description of the 'Login' Use Case

Use Case	Login
Description	Authenticates the player and loads their saved profile to continue gameplay.
Actor/s	Player
Pre-Condition	Player has an existing account; app is installed and launched.
Post-Condition	Player is authenticated, profile and saved progress loaded, access to Main Menu granted.
Basic Flow	• Player selects "Login" on welcome screen. • Player enters PIN code. • System loads player profile • System Player proceeds to Main Menu
Alternative Flow	• If invalid PIN code, system shows error and allows retry. • No account yet, system suggests Register.

Shown in Table 3.3 is the use case description of 'Login,' which outlines the player's interaction with the system to access their profile and continue gameplay.

Table 3.4 Tabular Description of the 'Register' Use Case

Use Case	Register
Description	Creates a new player account and initializes profile settings.
Actor/s	Player
Pre-Condition	Player has no existing account but app is installed and launched.
Post-Condition	New account created; profile initialized; access to Main Menu granted.
Basic Flow	• Player selects Register and enters required details. • Player proceeds to Main Menu
Alternative Flow	• Duplicate PIN code so system prompts for another. • Missing fields or invalid details; system shows a message.

Shown in Table 3.4 is the use case description of 'Register,' which outlines how a player creates a new account and initializes their profile to begin using the system.

Table 3.5 Tabular Description of the 'Play Game' Use Case

Use Case	Play Game
Description	Starts a gameplay session from the Main Menu
Actor/s	Player
Pre-Condition	Player is logged in and at Main Menu
Post-Condition	Game section initialized; player can select hero or continue the level.
Basic Flow	• Player taps Start Game • System loads to Select Hero panel; Player chooses Hero • Level Loads
Alternative Flow	N/A

Shown in Table 3.5 is the use case description of 'Play Game,' which outlines how the player initiates a gameplay session from the Main Menu and proceeds to hero selection and level loading.

Table 3.6 Tabular Description of the 'View Hero 3D model in AR' Use Case

Use Case	View Hero 3D model in AR
Description	Displays a 3D AR model of the selected hero for immersive learning.
Actor/s	Player
Pre-Condition	Player is in Hero archive; AR capable with camera permission.
Post-Condition	AR model rendered; player can exit AR mode.
Basic Flow	• Player taps View AR button • System Loads hero model. • Player interacts or exits.
Alternative Flow	N/A

Shown in Table 3.6 is the use case description of 'View Hero 3D model in AR,' which outlines how the player accesses and interacts with a hero's 3D augmented reality model for an immersive learning experience.

Table 3.7 Tabular Description of the 'Play Level' Use Case

Use Case	Play Level
Description	Player engages in gameplay, completing missions and challenges.
Actor/s	Player
Pre-Condition	Player selected a hero; level unlocked
Post-Condition	Level completed or not.
Basic Flow	• System loads level. • System shows How to Play. • Player interacts to the game. • Player collects coins and artifacts. • Completed or Game Over.
Alternative Flow	• Pause Game • Retry • Restart • Quit

Shown in Table 3.7 is the use case description of 'Play Level,' which outlines how the player engages in missions and challenges within a selected level, including interactions, coin and artifact collection, and possible outcomes such as completion or game over.

Table 3.8 Tabular Description of the 'Check Hero Unlock Status' Use Case

Use Case	Check Hero Unlock Status
Description	System evaluates conditions for unlocking heroes.
Actor/s	Player completes a level or opens Hero Selection.
Pre-Condition	The player had an active session and progress data stored in the database.
Post-Condition	The system displayed whether the selected hero was unlocked, locked, or available for purchase.
Basic Flow	• Player opened the Hero Selection screen. • The system retrieved player progress and artifact data from the database. • The system checked unlock conditions for each hero. • The system displayed the unlock status of the hero (unlocked, locked, or purchasable).
Alternative Flow	• Player attempted to access a locked hero. • The system prompted the player to complete required levels or collect specific artifacts. • If purchase was allowed, the system displayed the option to unlock the hero using coins or artifacts. • Player either continued gameplay to meet requirements or purchased the hero.

Shown in Table 3.8 is the use case description of 'Check Hero Unlock Status,' which outlines how the system evaluates player progress to determine whether heroes can be unlocked and displays their current status.

Table 3.9 Tabular Description of the 'Select Hero' Use Case

Use Case	Select Hero
Description	Player chooses a hero with distinct gameplay mechanics.
Actor/s	Player
Pre-Condition	Player authenticated; at least one hero unlocked.
Post-Condition	Selected hero loaded for next level.
Basic Flow	• Player opens Hero Selection. • System displays heroes. • Player chooses unlocked hero.
Alternative Flow	• Locked Hero tapped; don't have function.

Shown in Table 3.9 is the use case description of 'Select Hero,' which outlines how the player chooses an unlocked hero with distinct gameplay mechanics to be loaded for the next level.

Table 3.10 Tabular Description of the 'Collect Coin' Use Case

Use Case	Collect Coin
Description	Player collects coins during gameplay to unlock heroes or purchase items.
Actor/s	Player
Pre-Condition	Player is in an active level; coins available.
Post-Condition	Coins increased and updated.
Basic Flow	• Player collects coin after finishing the level.
Alternative Flow	• Game Over; coins don't persist.

Shown in Table 3.10 is the use case description of 'Collect Coin,' which outlines how the player gathers coins during gameplay to unlock heroes or purchase items, with updates reflected in their profile.

Table 3.11 Tabular Description of the 'Collect Artifact' Use Case

Use Case	Collect Artifact
Description	Player collects artifacts that unlock historical facts in the Hero Archive.
Actor/s	Player
Pre-Condition	Player is in an active level; artifact available.
Post-Condition	Artifact recorded; fact entry unlocked.
Basic Flow	• Player collects artifact after finishing the level. • System saves artifact to be viewed in Hero archive.
Alternative Flow	N/A

Shown in Table 3.11 is the use case description of 'Collect Artifact,' which outlines how the player gathers artifacts during gameplay that unlock historical facts in the Hero Archive for later viewing.

Table 3.12 Tabular Description of the 'Unlock & Purchase Hero' Use Case

Use Case	Unlock & Purchase Hero
Description	Player unlocks heroes via progress or purchases using coins.
Actor/s	Player
Pre-Condition	Player has required coins or completion status.
Post-Condition	Hero unlocked; coins deducted if purchased.
Basic Flow	• Player taps Unlock or Purchase button. • System checks eligibility. • Player confirms unlock. • System automatically unlocks after purchase or all level completion.
Alternative Flow	• Insufficient coins so system prevents purchase. • All levels not yet completed so system shows requirements.

Shown in Table 3.12 is the use case description of 'Unlock & Purchase Hero, which outlines how the player gains access to heroes either through gameplay progress or by spending coins, with the system validating eligibility and updating status accordingly.

UX Sitemap

Lakbay Bayani's UX Sitemap provided an overview of the system interface, showing how pages and features are connected. It served as a navigational guide to help players explore the game in an organized and user-friendly manner.

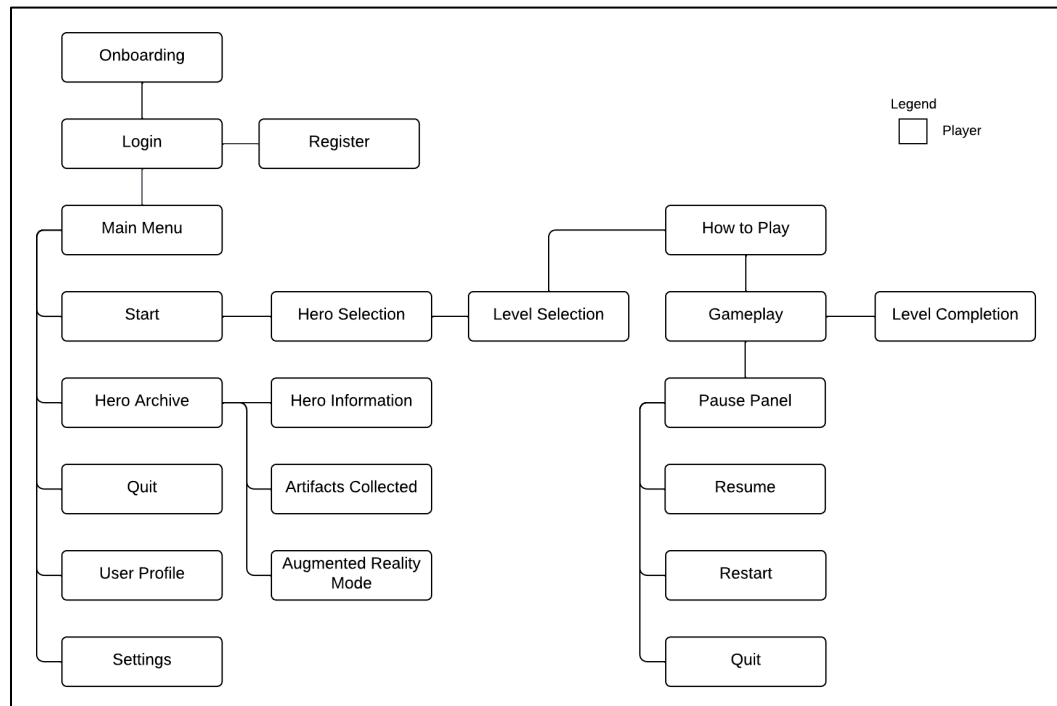


Figure 3.10 UX Sitemap

Illustrated in Figure 3.10 was the sitemap of Lakbay Bayani, which depicted the user interface navigation structure beginning with the onboarding process, Login and Register, followed by access to the Main Menu. From the main dashboard, players could navigate to key gameplay features such as Hero Selection, Level Selection, and Augmented Reality Mode, as well as supporting panels including Settings, User Profile, and Hero Archive. The sitemap also included in-game navigation elements such as the Pause Panel and Level

Completion screen. The diagram was presented without color coding, as the system was designed for a single user type, the player, eliminating the need for role-based differentiation. This structure ensured intuitive access to both educational and interactive content, supporting the game's goal of promoting Philippine heroism through immersive learning.

3.5 System Implementation

The implementation of Lakbay Bayani involved turning the system design into a working game. This process included writing code in C#, integrating modules within the Unity Engine, and testing the game on real Android devices to ensure compliance with requirements. Testing was conducted on Android 11 and above, specifically on AR Core-supported smartphones, to validate both the 2D platformer mechanics and the augmented reality features. Quality assurance focused on five attributes: functional suitability, performance efficiency, usability, reliability, and security. Once the game passed these checks, it was deployed as an APK build for installation on Android devices, allowing selected users to access and evaluate the system during pilot testing.

System Development

Lakbay Bayani was developed using a structured, modular approach to ensure that each feature could be implemented, tested, and integrated systematically. The following technologies, tools, and structures were utilized throughout the development process:

Unity Engine

Unity Engine is a cross-platform game development environment created by Unity Technologies. It is widely recognized for its ability to build interactive 2D, 3D, augmented reality (AR), and virtual reality (VR) applications. Developers use Unity because of its real-time rendering capabilities, integrated physics system, and extensive asset management tools, which make it suitable for both entertainment and educational projects (Unity Learn, 2025).

C# Programming Language

The Unity engine relies on C#, which was used to implement the game's logic, controls, and interactions. In Lakbay Bayani, C# scripts managed player movements, mission objectives, coin collection, and hero unlocking. It also handled the integration of AR features, ensuring smooth transitions between gameplay and augmented reality experiences. This programming backbone allowed developers to design responsive and scalable mechanics that matched the educational goals of the system (Team, 2025).

SQLite

SQLite was used as the database system for storing player progress, coin counts, and unlocked heroes. Its lightweight and serverless design made it ideal for mobile deployment, ensuring that data could be accessed quickly without requiring complex configurations. In Lakbay Bayani, SQLite tracked user achievements and saved game states, allowing players to continue their journey seamlessly across sessions (S, 2025).

AR Foundation

According to Adhav (2024) AR Foundation provided the framework for cross-platform AR development, supporting ARCore (Android). In the game, AR Foundation enabled features such as plane detection for placing heroes in real-world environments, image tracking for recognizing cultural symbols, and light estimation for realistic rendering. These features ensured that AR missions were interactive and consistent across devices, enhancing the educational immersion.

Ibis Paint

Ibis Paint was used to design and refine the digital assets of heroes and interactive elements. Its extensive brush library and manga-style tools allowed for detailed illustrations that matched the cultural and historical themes of the game. According to Noman (2025), Ibis paint and its variant Ibis Paint X offer major differences and benefits that make them versatile for creative projects. In Lakbay Bayani, Ibis Paint supported the creation of stylized hero portraits and background art, ensuring that visuals were both engaging and culturally authentic.

Photoshop

Adobe Photoshop was employed for professional editing and polishing of game assets. It was particularly useful for refining textures, adjusting colors, and preparing images for integration into Unity. In Lakbay Bayani, Photoshop ensured that hero models, interface elements, and promotional materials maintained a consistent and polished look, reinforcing the game's credibility as both an educational and entertainment platform (And, 2025).

Blender

Blender is a free and open-source 3D creation suite that supports modeling, rigging, animation, rendering, and visual effects (Barone, 2022). In Lakbay Bayani, Blender was used to design and animate three-dimensional hero models and cultural artifacts, ensuring characters were visually engaging and historically authentic. Its rigging and animation tools allowed heroes to move realistically during gameplay, while optimized rendering pipelines ensured smooth integration into Unity. By leveraging Blender's versatility, the project produced high-quality 3D assets at no cost, supporting its goal of combining cultural authenticity with interactive educational experiences.

System Testing

After developing Lakbay Bayani, the proponents conducted system testing to verify that the game functioned correctly, reliably, and in compliance with the intended requirements.

Technical Testing

Before presenting the game to users, the proponents performed technical testing to confirm that all features operated as intended and to identify potential errors. Three types of testing were conducted:

Unit Testing - Modules were tested including Player Movement, User Interface (UI), Augmented Reality (AR), Audio System, Reward System, Database, and Cutscene, to verify their correctness when executed independently. Each module was subjected to test cases derived from the system's functional requirements. The Player Movement module was examined to

ensure accurate responses to user inputs and proper collision detection. The UI module was tested for clarity, responsiveness, and accessibility of menus and navigation prompts. The AR module was validated for precise rendering and alignment of augmented elements without distortion.

The Audio System module was assessed for synchronization of background music and sound effects with gameplay events. The Reward System module was tested to confirm consistent awarding of points and badges. The Database module was evaluated for reliable storage and retrieval of hero information and player progress. Finally, the Cutscene module was tested to ensure smooth playback of narrative sequences. Results indicated that all modules performed correctly in isolation, with no critical errors detected, thereby establishing a stable foundation for subsequent integration testing.

Integration Testing - Integration testing was conducted to examine how modules interacted when combined into subsystems. Modules were tested including the Reward System with Database, the Cutscene with Audio System, and the UI with all other modules to ensure proper synchronization and navigation. The Reward System and Database confirmed accurate recording and retrieval of achievements, while the Cutscene and Audio System validated smooth alignment of narrative sequences with sound. The UI was tested across all modules to guarantee seamless transitions

between gameplay, rewards, and educational content. The AR module, though separate from gameplay, was tested independently to confirm stability and proper functioning when invoked.

System Testing - System testing was done to check the whole game and make sure all requirements were met. Modules were tested including User Interface (UI), Player Movement, Augmented Reality (AR), Audio System, Reward System, Database, and Cutscene. The UI module was checked for clear and easy navigation. The Player Movement module was tested to confirm smooth and responsive controls. The AR module, which works separately from gameplay, was tested to ensure stable performance and correct rendering when used. The Audio System module was reviewed to see if background music and sound effects matched the actions in the game. The Reward System and Database modules were tested together to confirm that achievements were saved and retrieved correctly. The Cutscene module was checked to make sure stories played smoothly without interrupting gameplay. Tests on real Android devices showed that all modules worked well together, with no major errors, proving the game is ready for users.

User Testing and Evaluation

Following technical testing, the proponents conducted User Acceptance Testing to assess usability, acceptability, and overall user satisfaction. Respondents were selected through quota sampling, ensuring

that each predefined group was properly represented. This method guaranteed that feedback was gathered from users who were most relevant to the game and its features.

A questionnaire aligned with the ISO/IEC 25010 software quality standard was prepared to collect user feedback. The instrument underwent expert validation by IT professionals and statistical validation by a licensed statistician, who ensured clarity, relevance, and accuracy of each item. Pilot testing results were analyzed, and recommendations were incorporated to refine the final questionnaire.

The finalized tool employed a 5-point Likert scale, ranging from Strongly Agree (5) to Strongly Disagree (1), to measure user satisfaction across various quality criteria. Responses collected through quota sampling were analyzed using mean scores and descriptive interpretation, with the statistician assisting in data processing to ensure accuracy and reliability.

System Deployment

The deployment of Lakbay Bayani was carried out in a mobile environment, specifically targeting Android devices. The finished system was packaged into an APK file, which served as the installation medium.

Deployment of Lakbay Bayani followed a step-by-step procedure to ensure that the system was properly installed and made operational. First, the APK was manually installed on Android devices, allowing the game to run offline without requiring internet connectivity. After installation, configuration was performed by

verifying the integration of the SQLite database to confirm that player progress, scores, and hero unlocks were stored and accessed correctly. Finally, account setup was completed during the onboarding process, where players were required to log in or register. This established their profile and enabled personalized tracking of progress throughout the game.

To assess stability, the proponents conducted pilot testing on a limited number of Android devices. This rollout ensured that performance, compatibility, and responsiveness were consistent across different hardware specifications. Alongside deployment, user training was provided in the form of a user manual and quick-start guide, which explained installation steps, gameplay mechanics, and troubleshooting procedures. These materials supported players in navigating the system and maximizing its educational features.

3.6 System Maintenance

After deployment, maintenance of Lakbay Bayani will be handled in a limited and practical manner. Since the game operates offline, most issues will be related to bugs, installation errors, or corrupted local data. Users will be asked to report any problems directly to the development team via email or messaging.

Reported issues will be logged, and critical errors will be addressed through minor updates or corrected APK files when possible. System updates will be minimal due to the offline nature of the game. Any adjustments or bug fixes will require users to install a new version of the APK manually. No automatic updates are provided.

Player progress and data are stored locally in the SQLite database. Periodic backups of the database can be done manually by users if desired. In case of data loss or

corruption, recovery can be performed by reinstalling the game and restoring any available backup files. Responsibility for maintenance will rest with the development team or other persons involved in the project for the immediate post-deployment period. This plan ensures that the game remains usable and playable offline while acknowledging limitations in long-term support.

CHAPTER IV

RESULTS AND DISCUSSIONS

This section presents the findings of the study in alignment with the specific objectives of the project. Each result is analyzed and interpreted to demonstrate how the project successfully addresses its goals.

1. To create (10) playable levels per hero, divided into three (3) easy, three (3) medium, and four (4) hard stages, ensuring increasing complexity and cognitive engagement.



Figure 4.1 Select Hero



Figure 4.2 10 Playable Levels

The system successfully implemented ten (10) playable levels for each hero. These levels were divided into three (3) easy, three (3) medium, and four (4) hard stages. This structure provided a gradual increase in difficulty, allowing players to progress from simple mechanics to more complex challenges. The division ensured that gameplay remained engaging while promoting cognitive development through problem-solving and strategic thinking.

2. **To assign each hero a specific gameplay genre such as action, puzzle-solving and narrative, and strategy that reflects their historical persona, and roles.**



Figure 4.3 Gameplay of Lapu-Lapu

The system successfully implemented Lapu-Lapu's melee combat gameplay. When playing, the user controls Lapu-Lapu with directional touch controls to move and attack enemies. Players swing weapons in close-range combat, block incoming strikes, and defeat waves of invaders. Early levels introduce basic attacks, while later stages add stronger enemies and boss battles, requiring timing and skill. This reflects his bravery in the Battle of Mactan, where combat was direct and physical.



Figure 4.4 Gameplay of Jose Rizal

The system successfully implemented José Rizal's puzzle-based gameplay, where players solve word and image puzzles that reflect his role as a reformist and intellectual. In each level, four related images are shown, and players must form a meaningful word using letter tiles within a time limit, earning coins for correct answers.

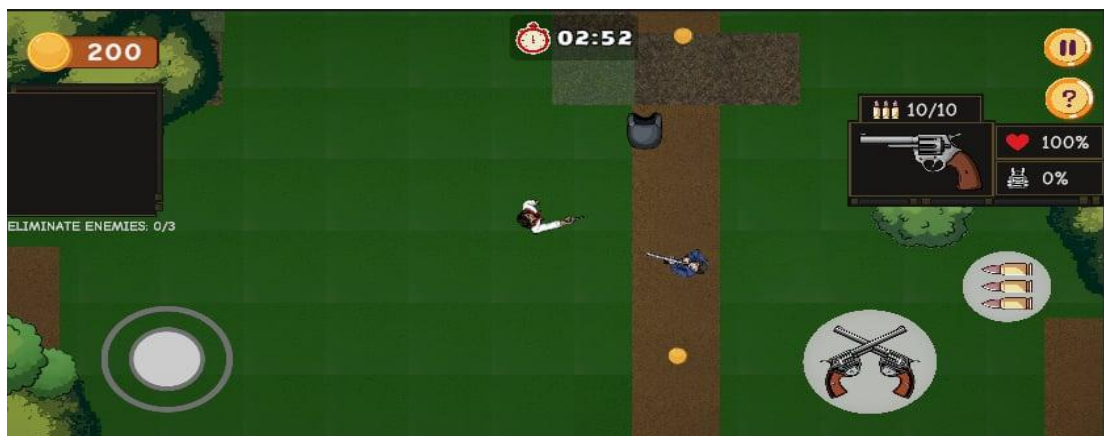


Figure 4.5 Gameplay of Andres Bonifacio

The system successfully implemented Andrés Bonifacio's gameplay as a top-down action shooter, reflecting his role as a revolutionary leader. Players guide Bonifacio across the battlefield, moving around obstacles while aiming and shooting at enemies to complete missions.



Figure 4.6 Gameplay of Emilio Aguinaldo

The system successfully implemented Emilio Aguinaldo's gameplay as a tower defense strategy. Players place defenses along a grid to stop waves of enemies, managing coins and upgrades as difficulty increases. This setup reflects Aguinaldo's role as a military leader, teaching players the value of planning and organized resistance.



Figure 4.7 Gameplay of Apolinario Mabini

The system successfully implemented Apolinario Mabini's gameplay as a word search puzzle, reflecting his role as a thinker and advisor. Players are presented with a grid of letters and must find hidden words related to revolutionary ideas and

governance. This fulfilled the objective of contextualizing educational content through interactive mechanics. It demonstrates that gameplay can serve as a medium for reinforcing cultural narratives, consistent with studies highlighting the value of experiential learning in history education.

3. To include narrative cutscenes at the beginning of each level to provide historical context, character background, and connection to events.

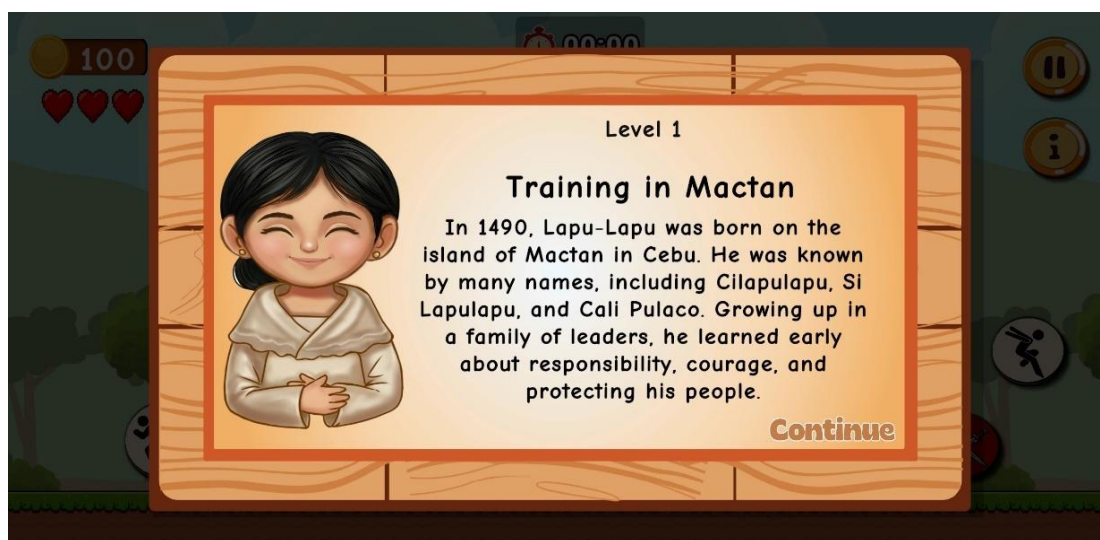


Figure 4.8 Cutscenes

Narrative cutscenes were successfully integrated at the start of each level. These sequences provided background information about the hero, their struggles, and the historical events they were part of. The inclusion of cutscenes ensured that gameplay was not isolated from its educational purpose but directly tied to historical storytelling. This fulfilled the objective of embedding historical context into the system. The approach aligns with findings in educational game design that narrative-driven learning enhances comprehension and engagement by situating tasks within meaningful contexts.

4. To develop a feature that displays a historical object or artifact related to the hero after each level is completed, along with its image and a brief description.



Figure 4.9 Artifact Collected

The system implemented an artifact display feature, presenting an image and description of a historical object after each level. This reinforced cultural learning by linking player achievements to tangible historical artifacts.

This fulfilled the objective of integrating educational content into gameplay. The feature validated the system's role as both an entertainment and learning tool, consistent with museum-based AR applications that combine interactivity with cultural preservation.

5. To design a hero and item information system that unlocks detailed profiles of national heroes along with encountered artifacts during gameplay in each level, providing historical context and significance upon level completion.



Figure 4.10 Hero's Information



Figure 4.11 Artifacts Collected

The system successfully provided a hero and item information archive. Upon level completion, players could access detailed profiles of heroes and artifacts, offering historical context and significance. This fulfilled the objective of extending learning beyond gameplay into archival-style exploration. It demonstrates that interactive systems can serve as repositories of cultural knowledge, supporting deeper engagement with history.

6. To implement a reward system that allows players to collect coins during gameplay and after completing game levels, which can be used to unlock at least one additional hero depending on the number of coins collected.

The system implemented a coin-based reward system, allowing players to collect coins during gameplay and use them to unlock additional heroes. This gamification strategy motivated continued play and exploration. It fulfilled the objective of sustaining engagement through rewards and aligns with gamification research, which emphasizes that reward mechanisms increase user satisfaction and encourage persistence in learning tasks.



Figure 4.12 Coins to Collect

This figure shows how coins appear during gameplay, either as drops from defeated enemies or as rewards for completing specific actions. Players are encouraged to actively engage with the game environment to gather coins, making the process interactive and rewarding.



Figure 4.13 Coins Collected

This figure displays the coin counter, which tracks the total coins earned by the player. It provides immediate feedback and a sense of progress, motivating players to continue playing in order to accumulate enough coins for future rewards.



Figure 4.14 Hero Purchasing

This figure illustrates how collected coins can be used to unlock additional heroes. By spending coins, players gain access to new characters with unique gameplay styles, reinforcing the connection between effort, reward, and exploration. This purchasing mechanic sustains long-term engagement by giving players clear goals to work toward.

7. To establish a hero unlocking system that allows players to unlock the next hero upon reaching level seven (7) with the previous hero, or alternatively purchase the hero using in-game coins.



Figure 4.15 Hero Unlocking

The system successfully implemented a hero unlocking mechanism. Players could unlock new heroes either by completing all ten levels of the previous hero or at least five levels, depending on historical connections. This fulfilled the objective of linking progression with cultural narratives. The design ensured continuity and encouraged players to explore multiple heroes, reinforcing the interconnectedness of Philippine history.

8. Integration of at least one AR interactive scene per hero, allowing players to view a 3D representation of the Filipino hero through their mobile devices.



Figure 4.16 Hero 3D Model in AR

The system integrated at least one AR interactive scene per hero, allowing players to view 3D representations through their mobile devices. This feature enhanced immersion and modernized historical learning. This fulfilled the objective of incorporating interactive technology into the system. The use of AR aligns with contemporary educational practices that leverage immersive tools to make learning more engaging and accessible.

Evaluation of the System

This section presents the quantitative analysis of user feedback on Lakbay Bayani using descriptive statistics. A total of 50 users participated in the evaluation through quota sampling, ensuring that the sample represented different groups relevant to the study. The evaluation was guided by the ISO/IEC 25010:2011 software quality model, which assessed the system's performance across five key quality attributes: Functional Suitability, Performance Efficiency, Reliability, Usability, and Security.

Each item was rated using a Likert scale, and the results were analyzed through weighted arithmetic means and standard deviations to determine the system's effectiveness and consistency. The use of quota sampling was intentional, as this method is widely applied in usability and educational technology studies to guarantee balanced representation across categories such as, students, educators, and casual gamers. By setting quotas, the researchers ensured that no single group dominated the evaluation, making the findings more reliable and reflective of diverse perspectives

Table 4.1 User Evaluation Results

1. Functional Suitability	Mean	SD	Descriptive Interpretation
1.1 Functional Completeness: The game allows players to select heroes, play missions, collect coins, and access the Hero Archive panel as intended.	4.34	0.52	Agree
1.2 Functional Correctness: Heroes unlock only after completing 10 required levels or using earned coins to purchase as designed.	4.22	0.55	Agree

1.3 Functional Correctness: The Hero Archive displays accurate, complete, and historically aligned information for every hero.	4.36	0.66	Agree
1.4 Functional Appropriateness: The game promotes historical awareness and Filipino heroism through interactive gameplay.	4.34	0.59	Agree
2. Performance Efficiency			
2.1 Time Behavior: Levels and missions load within 1-3 seconds, ensuring smooth gameplay without noticeable delay.	4.10	0.71	Agree
2.2 Time behavior: Interactive elements such as buttons, menus, and non-AR features respond immediately when tapped.	4.06	0.68	Agree
2.3 Time behavior: AR features fully activate within 3–6 seconds, which remains within an acceptable loading time for uninterrupted gameplay.	4.16	0.62	Agree
2.4 Capacity: The game can run smoothly without using too much device memory or storage.	4.14	0.70	Agree
3. Reliability			
3.1 Maturity: The game operates consistently without crashing during normal gameplay.	4.34	0.63	Agree
3.2 Availability: The game is accessible and playable whenever the player opens it, without unexpected downtime.	4.22	0.62	Agree
3.3 Fault Tolerance: The game continues to function properly even when minor errors occur such as lag spikes, missing assets, or interrupted AR embedding.	4.42	0.57	Agree

3.4 Fault Tolerance If the game closes unexpectedly, the system restarts and returns to the loading screen, requiring the player to log in again before starting the level.	4.34	0.56	Agree
4. Usability			
4.1 Operability: Controls for movement, jumping, attacking, interacting with the heroes and activating AR are intuitive and easy to use.	4.34	0.66	Agree
4.2 Appropriateness Recognizability: The interface clearly reflects that the game is educational and focused on Philippine heroism.	4.30	0.65	Agree
4.3 Learnability: It is easy to learn how to play the game without needing additional instructions.	4.16	0.62	Agree
4.4 User Interface Aesthetics: Text, menus, icons, audio, and UI elements are visually clear, consistent, and readable and contents are understandable.	4.04	0.57	Agree
4.5 Accessibility: Players can comfortably navigate missions, levels, AR features, and the Hero Archive.	4.28	0.67	Agree
5. Security			
5.1 Authenticity: The game lets users make an account and use a PIN so they can log in quickly the next time by just entering the PIN.	3.80	0.57	Agree
5.2 Confidentiality: Each player has a unique PIN that protects their account and allows them to access or retrieve their saved game data.	3.76	0.66	Agree
5.3 Integrity: Player accounts can only use unique pin code that securely stored to prevent unauthorized use and duplicate user.	3.90	0.54	Agree

Table 4.2 Overall Quality Evaluation of the Lakbay Bayani

Indicators	Mean	SD	Descriptive Interpretation
Functional Suitability	4.32	0.37	Functional
Performance Efficiency	4.12	0.47	Efficient
Reliability	4.33	0.35	Reliable
Usability	4.22	0.41	Usable
Security	3.82	0.41	Secured
Overall Mean	4.16	0.29	With High Quality

Table 4.1 presents the overall quality evaluation of Lakbay Bayani based on the ISO/IEC 25010 software quality model. The indicator Functional Suitability received a weighted mean of 4.32, interpreted as Functional, with a standard deviation of 0.37, indicating strong agreement among respondents. Performance Efficiency scored 4.12, interpreted as Efficient, with a standard deviation of 0.47, suggesting slight variation in user experience across devices. Reliability obtained a mean of 4.33, interpreted as Reliable, and a standard deviation of 0.35, showing consistent feedback on system stability. Usability was rated at 4.22, interpreted as Usable, with a standard deviation of 0.41, reflecting general ease of use and accessibility. Security received the lowest mean of 3.82, interpreted as Secured, with a standard deviation of 0.41, indicating agreement but also highlighting potential areas for improvement. Overall, the system achieved a mean score of 4.16, interpreted as With High Quality, and a low standard deviation of 0.29, confirming consistent user satisfaction and readiness for deployment.

CHAPTER VI

SUMMARY, CONCLUSIONS, AND RECOMMENDATIONS

5.1 Summary

The purpose of this project was to design, develop, and evaluate Lakbay Bayani, a mobile educational game that integrates interactive mechanics and augmented reality to promote Philippine history and culture. The project was undertaken to address the need for innovative and engaging tools that make historical learning more accessible and enjoyable for students and the general public.

The system was designed with structured gameplay, including ten levels per hero, distinct genres, and narrative cutscenes to provide historical context. Interactive features such as artifact displays, hero archives, reward systems, unlocking mechanisms, and AR scenes were incorporated to enhance user engagement. The methodology involved iterative design, system refinement, and user evaluation guided by the ISO/IEC 25010:2011 software quality model.

Playtesting sessions were conducted with a diverse group of users including casual players, students, and educators, and feedback was collected through questionnaires. The evaluation confirmed that the system is functional, efficient, reliable, usable, and secured, with an overall weighted mean of 4.16, interpreted as With High Quality. These findings validate that Lakbay Bayani adheres to recognized software quality standards and demonstrates high quality as an educational application that effectively integrates interactive learning with the promotion of Philippine culture.

5.2 Conclusions

Based on the findings of the study, it can be concluded that Lakbay Bayani successfully achieved its objectives of designing, developing, and evaluating an educational mobile game that promotes Philippine history and culture. The system was verified to be functional, efficient, reliable, usable, and secured, as confirmed through user evaluation guided by the ISO/IEC 25010:2011 standards. Respondents consistently agreed that the game delivered its intended features, operated smoothly, and provided an engaging and accessible learning experience.

The evaluation results further indicated that the integration of interactive mechanics and augmented reality enhanced user engagement and contributed significantly to the educational effectiveness of the system. The overall weighted mean of 4.16, interpreted as With High Quality, validated that the application met international software quality standards and was ready for deployment.

Moreover, the project demonstrated that technology-based solutions, particularly mobile applications, can play a vital role in making historical learning more interactive, enjoyable, and culturally relevant. By combining entertainment with education, Lakbay Bayani provided a practical and innovative tool for promoting Philippine heroism while supporting modern approaches to teaching and learning.

5.3 Recommendations

Based on the evaluation results and user feedback, several recommendations were identified to further improve Lakbay Bayani's effectiveness, usability, and educational impact. Users recognized the game as a valuable tool for promoting Philippine heroism, particularly for learners who benefit from interactive and engaging approaches

to history. The structured levels and augmented reality features were highlighted as effective mechanisms for combining entertainment with education, allowing players to learn about national heroes while actively participating in gameplay.

To enhance inclusivity, the addition of a language selection feature through a dropdown menu or button was recommended, enabling players to choose their preferred language content. This would make the game more accessible to a wider audience and ensure that cultural and historical lessons could be understood by learners from diverse linguistic backgrounds. Expanding the game content to include additional levels, challenges, and interactive activities was also suggested to increase engagement and provide deeper learning experiences.

Another recommendation is to add a multiplayer feature so players can play together or against each other. This would make the game more fun and interactive, as it encourages teamwork, friendly competition, and social interaction.

Users further recommended optimizing the system for deployment across multiple platforms, such as iOS and desktop environments, to broaden accessibility. Strengthening database security and refining AR interactions were also identified as important steps to ensure reliable performance and protect player data.

Implementing these suggestions would strengthen Lakbay Bayani as a sustainable, educational, and user-friendly platform, while continuing to promote Philippine culture, history, and heroism through interactive learning and augmented reality.

LITERATURE CITED

- Admin. (2024, July 20). *VP Jocson examines changing narratives of Filipino heroism in 2nd lecture as PUP professorial chairholder in literature ed.* Research Institute for Teacher Quality (RITQ). <https://www.ritq.ph/?p=4115>
- Adhav, R. (2024, December 27). *Google AR: A Comprehensive Guide to ARCORE.* Designial- Product Development & Marketing | Silicon Valley, California. <https://designial.com/blogs/google-ar-a-comprehensive-guide-to-arcore/>
- Albano, G., Arzarello, F., & Iacono, U. D. (2020). Digital Inquiry Through Games. *Technology Knowledge and Learning*, 26(3), 577–595. <https://doi.org/10.1007/s10758-020-09459-1>
- Alduwairej, M. A. (2020). *Digital game based learning (A literature review).* University of Bisha. <https://www.ajrsp.com/en/Archive/issue-66/2>
- Alotaibi, M. S. (2024). Game-based learning in early childhood education: a systematic review and meta-analysis. *Frontiers in Psychology*, 15. <https://doi.org/10.3389/fpsyg.2024.1307881>
- And, S. (2025). *What is Photoshop? History, features, and use cases| AND Academy.* AND Academy. <https://www.andacademy.com/resources/blog/graphic-design/what-is-photoshop/>
- Andres, R. B. (2024). Heroism from the eyes of the young generation. Friedrich Naumann Foundation Philippines. <https://www.freiheit.org/philippines/heroism-eyes-young-generation>
- Barone, R. (2022, April 11). *What is Blender software? What is it used for? 3D models & more.* iD Tech.
- Bellingham, H. (2022, May 31). Award-winning puzzle game Monument Valley is heading to PC in a new collection. *GamesRadar+*. <https://www.gamesradar.com/award-winning-puzzle-game-monument-valley-is-heading-to-pc-in-a-new-collection/>
- Calipay, C. (2024). *Albay town youth leaders educate kids on lives of heroes.* Philippine News Agency. <https://www.pna.gov.ph/index.php/articles/1231888>
- Centeno-Martín, H., Toledano-Buendía, S., & Ardévol-Abreu, A. (2023). Who interacts with communicative AI and what attitudes are displayed toward it? Sociodemographic, personality, and futurephobia variables. *Anuario ThinkEPI*. <https://doi.org/10.3145/epi.2023.sep.02>
- Dhaas, A. (2024). Augmented Reality in Education: A Review of Learning Outcomes and Pedagogical Implications. *American Journal of Computing and Engineering*. 7. 1-18. [10.47672/ajce.2028](https://doi.org/10.47672/ajce.2028).

- Ebro, E. (2022). KasaySAYAn: Revisiting History among 21st Century Learners. *The Normal Lights*, 16(1). <https://doi.org/10.56278/tnl.v16i1.1709>
- Earichappan, D., & Hashim, H. (2024). Game-Based Learning in ESL Context: A Systematic Literature Review (2020-2024). *International Journal of Academic Research in Progressive Education and Development*, 13(3). <https://doi.org/10.6007/ijarped/v13-i3/22895>
- Freese, P. (2021). *Pokémon GO: All events in October 2021 and which are worth it*. Mein-MMO. <https://mein-mmo.de/en/pokemon-go-all-events-in-october-2021-and-which-are-worth-it,726019/>
- García-Martínez, I., Fernández-Batanero, J. M., Fernández-Cerero, J., & León, S. P. (2023). Analysing the Impact of artificial intelligence and computational sciences on student performance: systematic review and meta-analysis. *Journal of New Approaches in Educational Research*, 12(1), 171–197. <https://doi.org/10.7821/naer.2023.1.1240>
- Garcia, Manuel. (2020). Augmented Reality in History Education: An Immersive Storytelling of American Colonisation Period in the Philippines. *International Journal of Learning Technology*, 15, 234–254. https://www.researchgate.net/publication/348253330_Augmented_Reality_in_History_Education_An_Immersive_Storytelling_of_American_Colonisation_Period_in_the_Philippines
- Haron, H. @ N., Masrom, M., Ya'acob, S., & Sabri, S. A. (2021). The Challenges and Constraints of Online Teaching and Learning in the New Normal Environment. https://hrmars.com/papers_submitted/9825/the-challenges-and-constraints-of-online-teaching-and-learning-in-the-new-normal-environment.pdf
- Hidayat, R., Qi, T. Y., Ariffin, P. N. B. T., Hadzri, M. H. B. M., Chin, L. M., Ning, J. L. X., & Nasir, N. (2024). Online game-based learning in mathematics education among Generation Z: A systematic review. *International Electronic Journal of Mathematics Education*, 19(1), em0763. <https://doi.org/10.29333/iejme/xxxx>
- Hui, H. B., & Mahmud, M. S. (2023). Influence of game-based learning in mathematics education on the students' cognitive and affective domain: A systematic review. *Frontiers in Psychology*, 14. <https://doi.org/10.3389/fpsyg.2023.1105806>
- Jitsuuuuu7. (2022, May 12). *Hi guys! We are IT students and we created an educational game about Philippine history and heroes* [Online forum post]. Reddit. https://www.reddit.com/r/Philippines/comments/utksmu/hi_guys_we_are_it_students_and_we_created_an/
- Kelleher, L. (2025, February 6). *Defining the Platformer genre — Kelleher Bros*. Kelleher Bros. <https://www.kelleherbros.com/blog/2024/3/4/defining-the-platformer-genre>

- KendiKorp. (2022). *Galà*. A cultural mobile game showcasing Filipino landmarks and traditions. <https://kendikorp.com/gala/>
- Lee, B. N. (2023). DIGITAL TOOLS & INQUIRY-BASED LEARNING IN HISTORY EDUCATION. *Muallim Journal of Social Science and Humanities*, 78–88. <https://doi.org/10.33306/mjssh/255>
- Magazine, A. (2023). *Ayala Museum reveals long-lost Juan Luna Masterpiece* — Art+ Magazine. Art+ Magazine. <https://artplus.ph/features/ayala-museum-reveals-long-lost-juan-luna-masterpiece>
- Martins, J. (2025, February 1). SCRum: The Most Popular Agile Framework [2025] • Asana. *Asana*. <https://asana.com/resources/what-is-scrum>
- Mantikayan, J. M., Abdulgani, M. A., Mohamad, H., & Sinsuat, D. R. R. (2022b). The Effectiveness of Augmented Reality (AR) as a tool of office for Ministry of Basic, Higher, and Technical Education in Bangsamoro Autonomous Region in Muslim Mindanao. <https://ejournals.ph/article.php?id=20363>
- Montemayor, M. T. (2024). NHCP codifies criteria for ‘hero’. <https://www.pna.gov.ph/index.php/articles/1221070>
- Nadeem, M., Oroszlanyova, M., & Farag, W. (2023). Effect of Digital Game-Based Learning on student engagement and motivation. *Computers*, 12(9), 177. <https://doi.org/10.3390/computers12090177>
- Noman, F. (2025, December 4). *Ibis Paint vs Ibis Paint X – Major Differences and Benefits*. Khired Kids. <https://khiredkids.com/ibis-paint-vs-ibis-paint-x/>
- Padli, I. N. W., & Yasin, R. M. (2023). Challenges of Technology Integration in Teaching History and its Relationship with Student Motivation. *International Journal of Academic Research in Progressive Education and Development*, 12(4). <https://doi.org/10.6007/ijarped/v12-i4/19537>
- Pratama, Pandu & Amalia, Mufidah & Zamahsari, Gamal & Pangestu, Gusti. (2022). Design and Development Bineka: A Platformer 2D Game for Learning Indonesian Language. 2537-2543. 10.46254/EU05.20220493. https://www.researchgate.net/publication/372284737_Design_and_Development_Bineka_A_Platformer_2D_Game_for_Learning_Indonesian_Language
- Morales, A. R., & Regio, V. R. (2024). *Augmented Reality-Based Learning Aid (AR-BLA) in enhancing the Grade 7 students’ learning performance in Biology at Guinayangan National High School, Guinayangan District, Division of Quezon. Psychology and Education: A Multidisciplinary Journal*
- Sinha, S. (2023, June 6). *Augmented Reality (AR) in Education: a staggering insight into the future*. eLearning Industry. <https://elearningindustry.com/augmented-reality-in-education-staggering-insight-into-future>

- S, R. A. (2025, September 10). *What is SQLite? Everything You Need to Know*. Simplilearn.com. <https://www.simplilearn.com/tutorials/sql-tutorial/what-is-sqlite>
- Tamosevicius, R. (2022, November 22). *Why is Game-Based learning important?* eLearning Industry. <https://elearningindustry.com/why-is-game-based-learning-important>
- Team, P. C. (2025, March 20). What is C# Programming? A Beginner's Guide. *Pluralsight*. <https://www.pluralsight.com/resources/blog/software-development/everything-you-need-to-know-about-c->
- Unity learn*. (2025). Unity Learn. <https://learn.unity.com/tutorial/what-is-unity>
- Vivas, J. (2020, August 31). *Filipino fighting game 'Bayani' is historic!* Manila Bulletin. <https://mb.com.ph/2020/08/31/filipino-fighting-game-bayani-is-historic/>
- Wang, L., Chen, B., Hwang, G., Guan, J., & Wang, Y. (2022). Effects of digital game-based STEM education on students' learning achievement: a meta-analysis. *International Journal of STEM Education*, 9(1). <https://doi.org/10.1186/s40594-022-00344-0>
- Wejebe, A. (2019, May 20). *Ranida Games Discusses Bayani: Kanino Ka Kakampi?! at IndieCade 2018 - Cliqist*. Cliqist. <https://cliqist.com/2018/10/29/ranida-games-discusses-bayani-kanino-ka-kakampi-at-indiecade-2018/#>

APPENDICES

Letter of Request and Approval to Conduct the Study

Routing Slip

Expert Evaluation Form

Survey User Evaluation Questionnaire

Sample filled-out User Evaluation Questionnaires

Consolidated User Evaluation Results

User Testing Photo Documentation

Curriculum Vitae

ROSHELLE MASANGUID

Purok D.A., Barangay Matiao

City of Mati, Davao Oriental

+639553523895

roshellemasanguid@gmail.com

**PERSONAL DATA**

Date of Birth	:	February 13, 2004
Age	:	21
Place of Birth	:	Barangay Rizal, Manay, Davao Oriental
Sex	:	Female
Height	:	5'0 ft
Weight	:	47 kg
Civil Status	:	Single
Nationality	:	Filipino

SKILLS AND INTEREST

- ✓ Video Editing
- ✓ Reading
- ✓ Watching Movies
- ✓ Hiking

EDUCATIONAL BACKGROUND

TERTIARY	Bachelor of Science in Information Technology Davao Oriental State University Guang-guang, Dahican, City of Mati, Davao Oriental 2022-2026
-----------------	---

SECONDARY	General Academic Strand Matiao National High School Archie Heights, Matiao, City of Mati 2016-2022
------------------	---

ELEMENTARY	Jose Bitac Elementary School Rizal, Manay, Davao Oriental 2010-2016
-------------------	---

NICHOLE L. MANGOPOT

Central, Manay, Davao Oriental

+639774154430

nicholemangopot7@gmail.com**PERSONAL DATA**

Date of Birth : September 10, 2004
Age : 21
Place of Birth : Manay, Davao Oriental
Sex : Female
Height : 5'5 ft
Weight : 59 kg
Civil Status : Single
Nationality : Filipino

SKILLS AND INTEREST

- ✓ Programming
- ✓ Game Designing
- ✓ Adaptability
- ✓ Teamwork

EDUCATIONAL BACKGROUND

TERTIARY Bachelor of Science in Information Technology
Davao Oriental State University
Guang-guang, Dahican, City of Mati, Davao Oriental

2022-2026

SECONDARY General Academic Strand
Evaristo Moralizon National Vocational High School
Central, Manay, Davao Oriental

2016-2022

ELEMENTARY Gregorio Moralizon Elementary School 1
Central, Manay, Davao Oriental

2010-2016

CHRISTIAN BACUS

Barangay Matiao, Central
City of Mati, Davao Oriental
+639156251390
christianbacus77@gmail.com

**PERSONAL DATA**

Date of Birth	:	March 31, 2004
Age	:	21
Place of Birth	:	Manay, Davao Oriental
Sex	:	Male
Height	:	5'6 ft
Weight	:	59 kg
Civil Status	:	Single
Nationality	:	Filipino

SKILLS AND INTEREST

- ✓ Programming
- ✓ Game Developing
- ✓ Music / Playing Instruments
- ✓ Cooking

EDUCATIONAL BACKGROUND

TERTIARY	Bachelor of Science in Information Technology Davao Oriental State University Guang-guang, Dahican, City of Mati, Davao Oriental 2022-2026
SECONDARY	TVL Mati Doctors Academy National Highway, Dahican, Mati City, Davao Oriental 2016-2022
ELEMENTARY	Matiao Central Elementary School Matiao, City of Mati, Davao Oriental 2010-2016